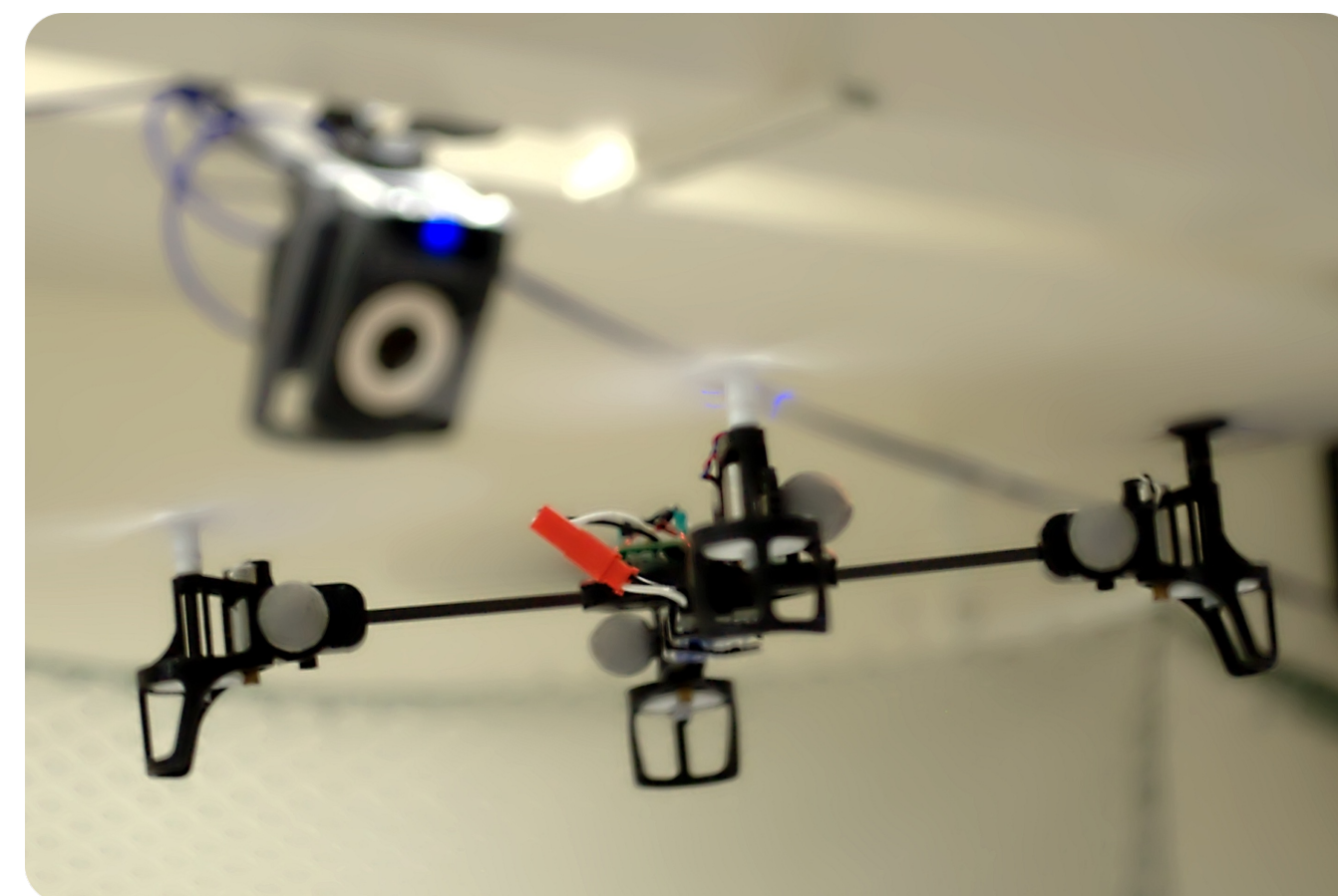


Dynamic modeling and motion control of quadrotor UAVs



Rita Cunha / J.R. Azinheira

Lisboa, Setembro 2022

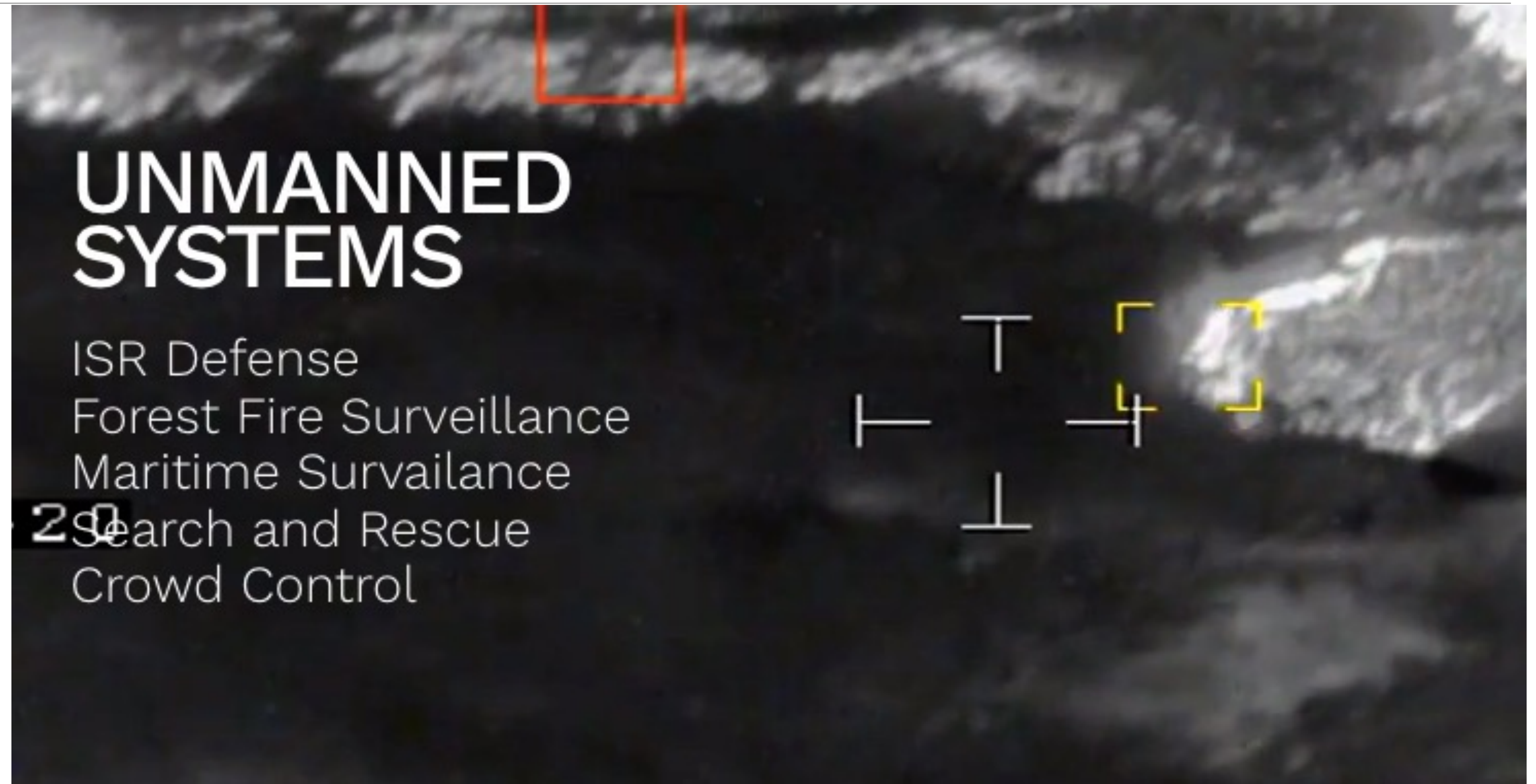


UAV applications ?

→ Portuguese manufacturers

Aerial inspection, Precision Agriculture, Entertainment,....

And
Military
applications



Portuguese UAV manufacturer example (1)

- <https://www.uavision.com>



Portuguese UAV manufacturer example (2)

- <http://uas.tekever.com/>

TEKEVER AR4



TEKEVER AR5



Portuguese UAV manufacturer example (3)

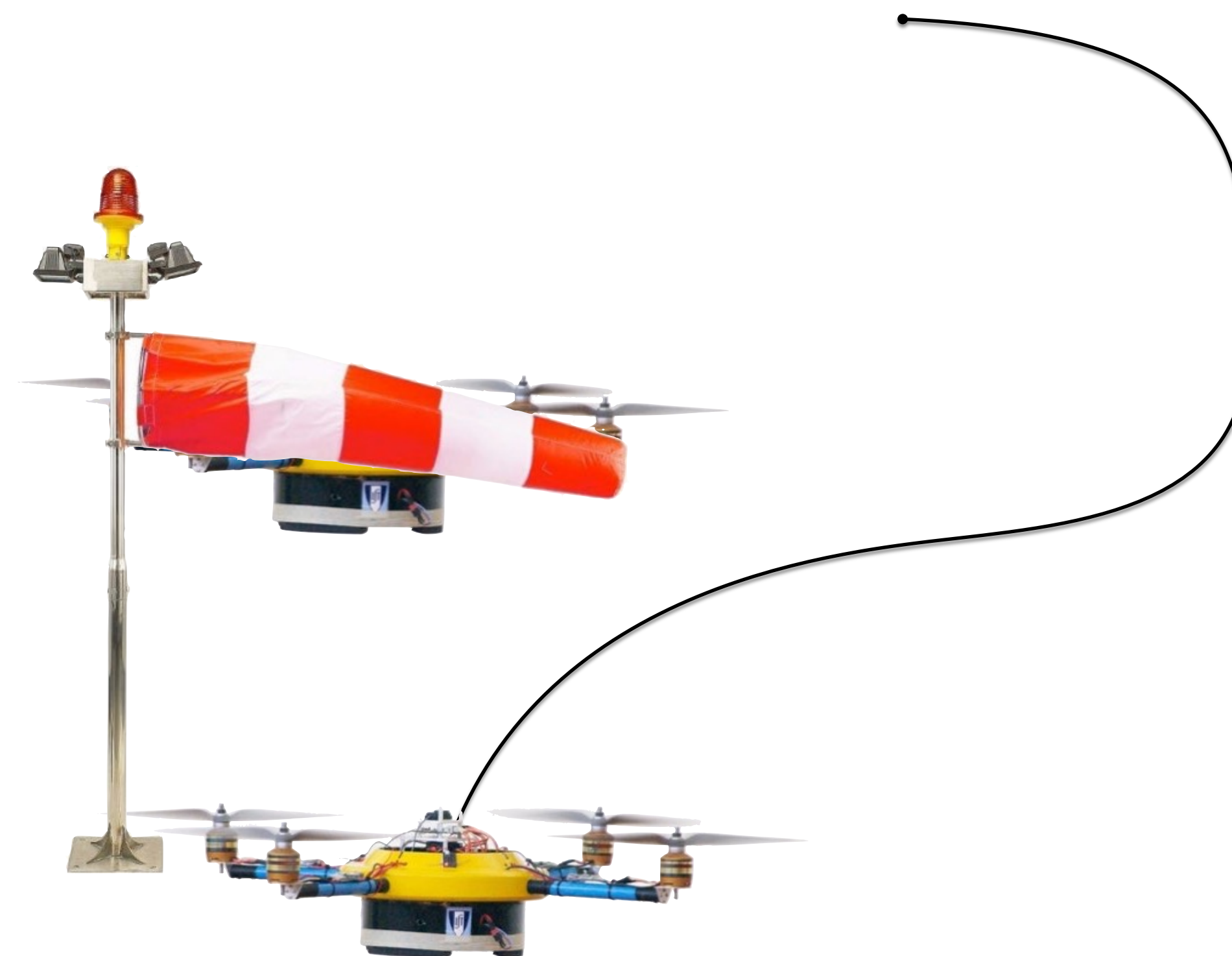
- <http://www.spinworks.pt/unmanned-systems/>



Motion Control of Aerial Vehicles

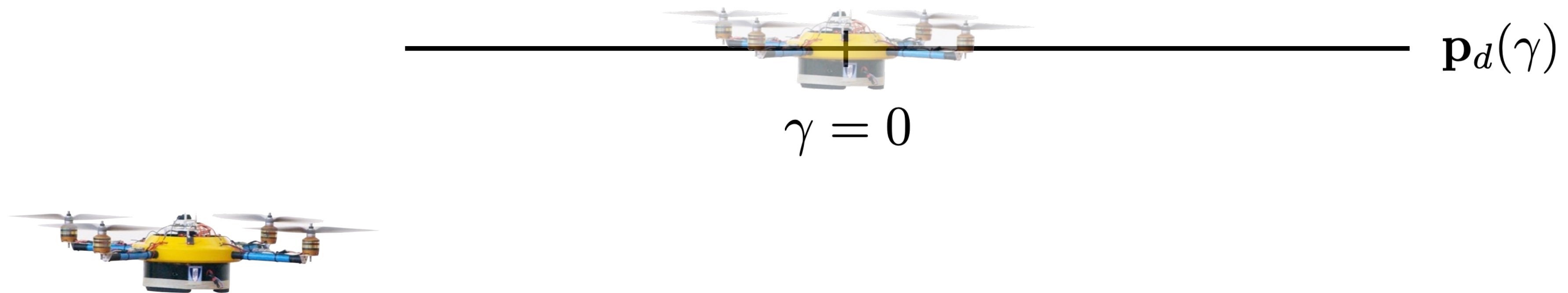
Trajectory Tracking - Control Objectives

- Track a trajectory
- Realistic model
- Robustness to disturbances or model uncertainties
- Actuation bounds
- Large basin of attraction



Path Following Control

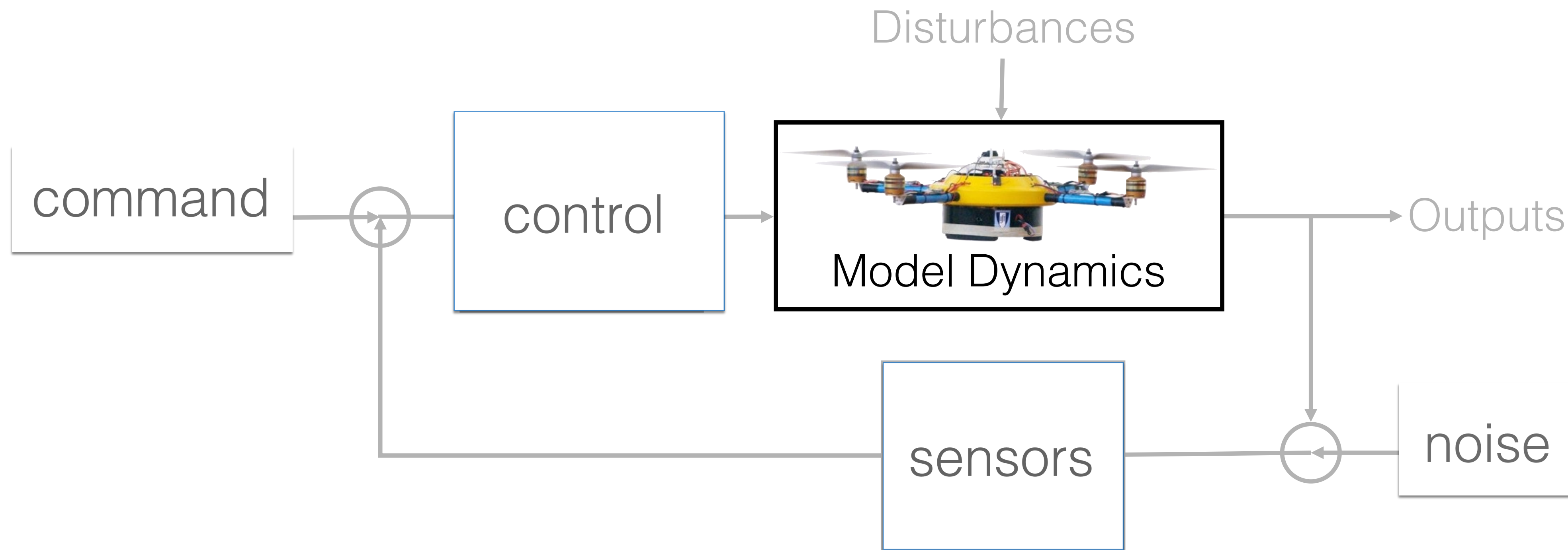
- Path Following vs. Trajectory Tracking



Far: Ensure control law is well defined

Near: Velocity control

The quadrotor as a control system



Let's start with the model

The quadrotor as a rigid body

Kinematics

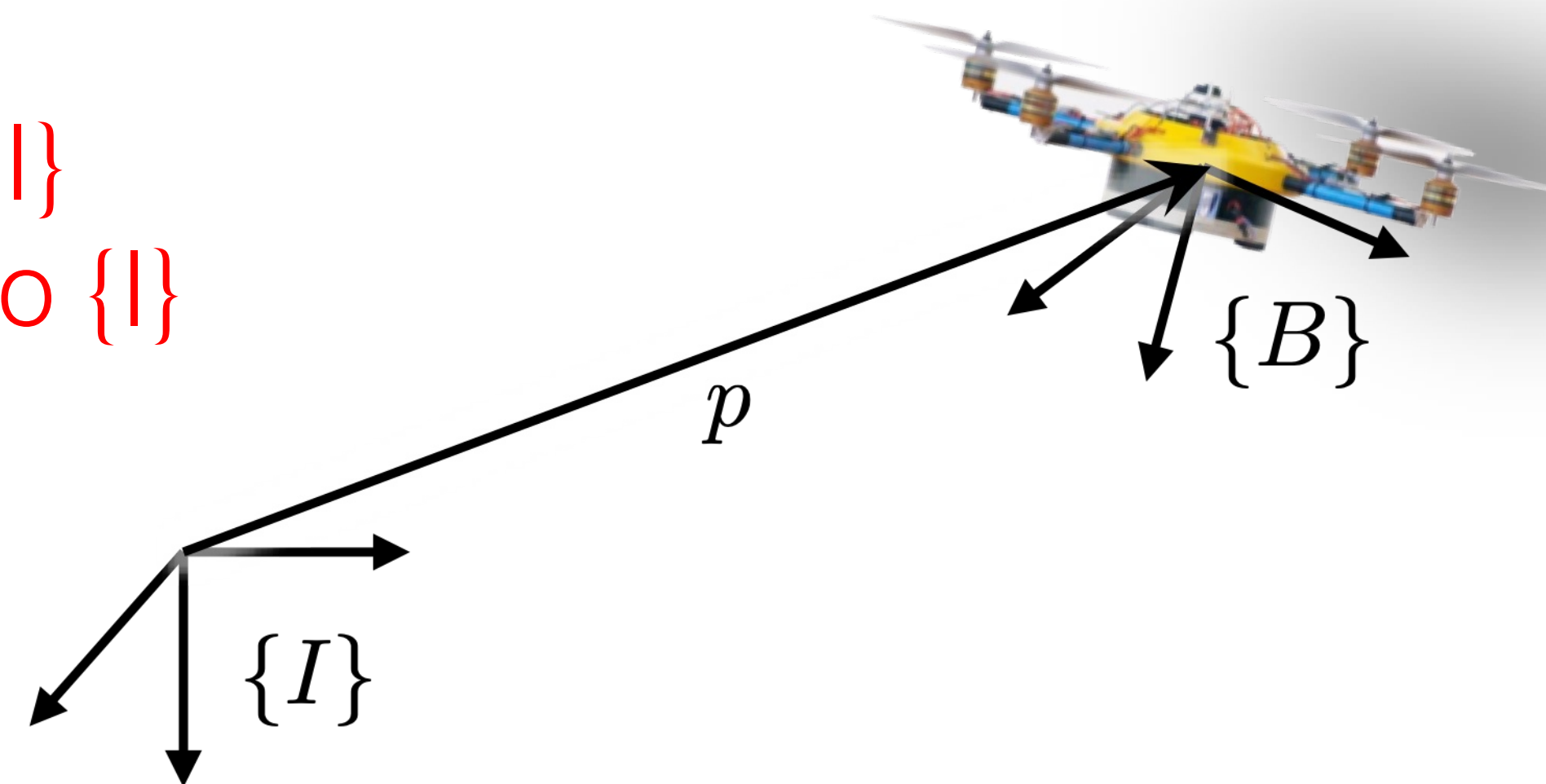
$$\begin{cases} \dot{p} = v \\ \dot{R} = RS(\omega) \end{cases}$$

Velocity in $\{I\}$
 R from $\{B\}$ to $\{I\}$

Dynamics

$$\begin{cases} m\dot{v} = f \\ J\dot{\omega} = -S(\omega)J\omega + n \end{cases}$$

$$S(\omega)a = \omega \times a, \text{ for } \omega, a \in \mathbb{R}^3$$



$m \in \mathbb{R}$ – mass

$J \in \mathbb{R}^{3 \times 3}$ – Tensor of inertia expressed in $\{B\}$

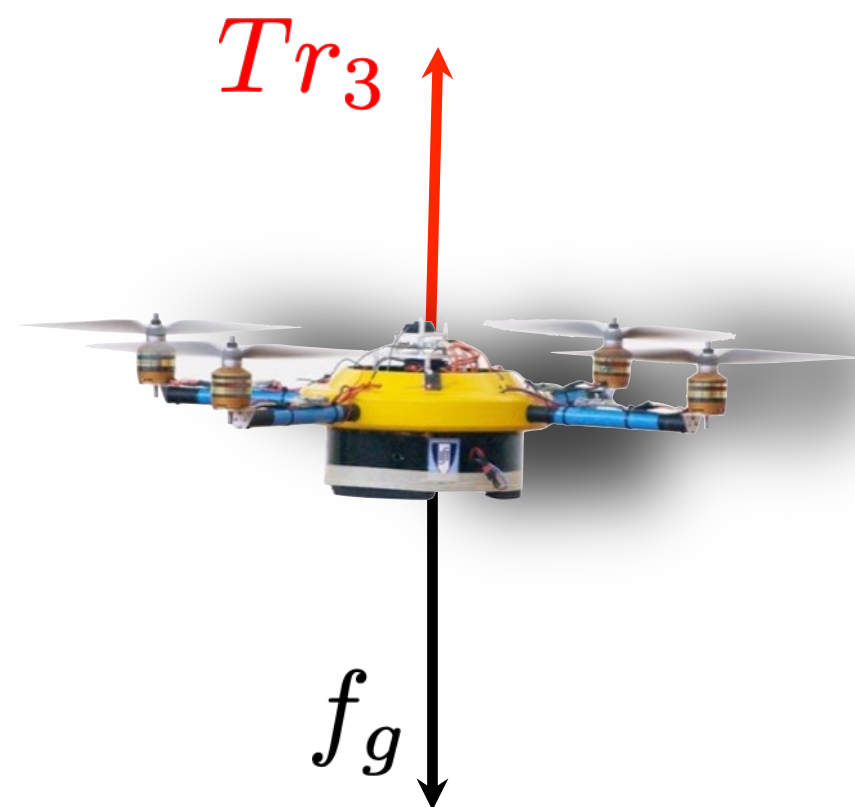
$f \in \mathbb{R}^3$ – External forces expressed in $\{I\}$

$n \in \mathbb{R}^3$ – External moments expressed in $\{B\}$

Quadrotor forces

Underactuated system: 12 states, 4 inputs

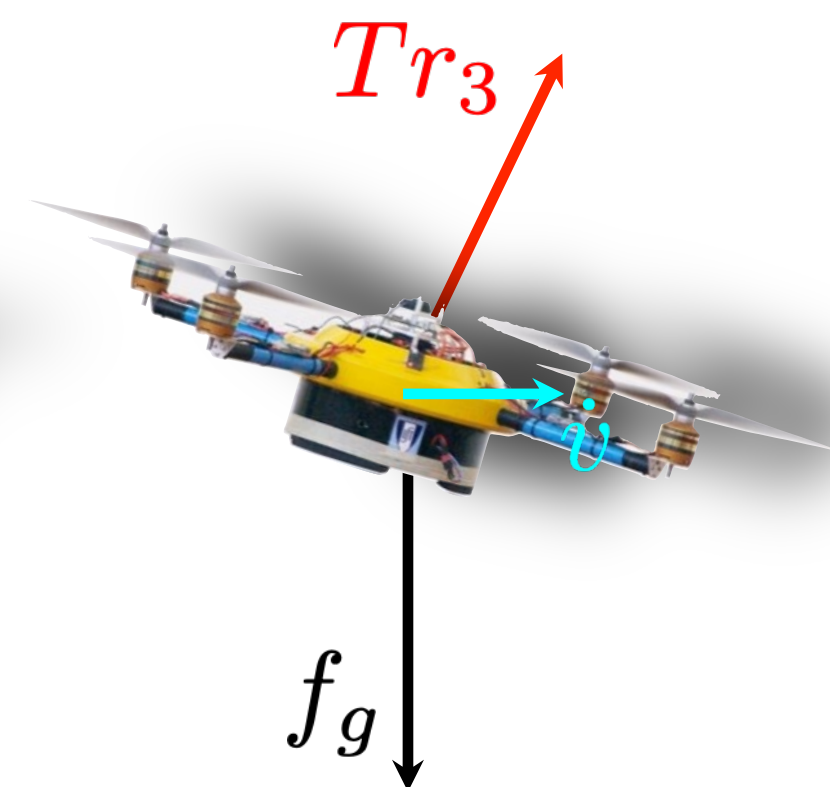
At rest (hover)



$$v = 0$$

$$Tr_3 = f_g$$

Accelerating



$$\dot{v} \neq 0$$

$$Tr_3 = f_g - m\dot{v}$$

Control objective:

Steer the position to a desired trajectory

$$p(t) \rightarrow p_d(t)$$

using (T, n) as inputs.

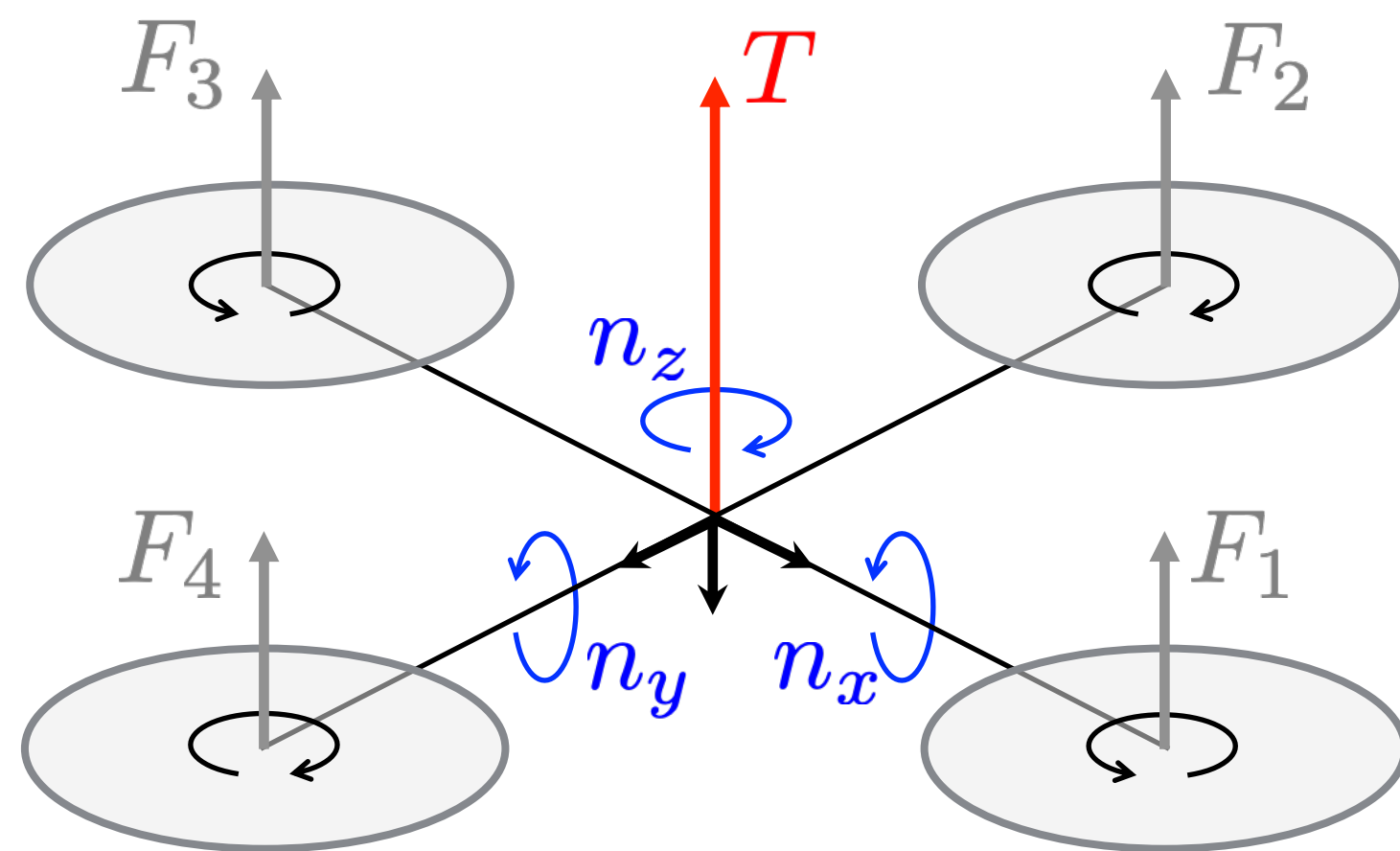
Can only prescribe 4 states:

e.g. 3D position and rotation about z-axis (yaw)

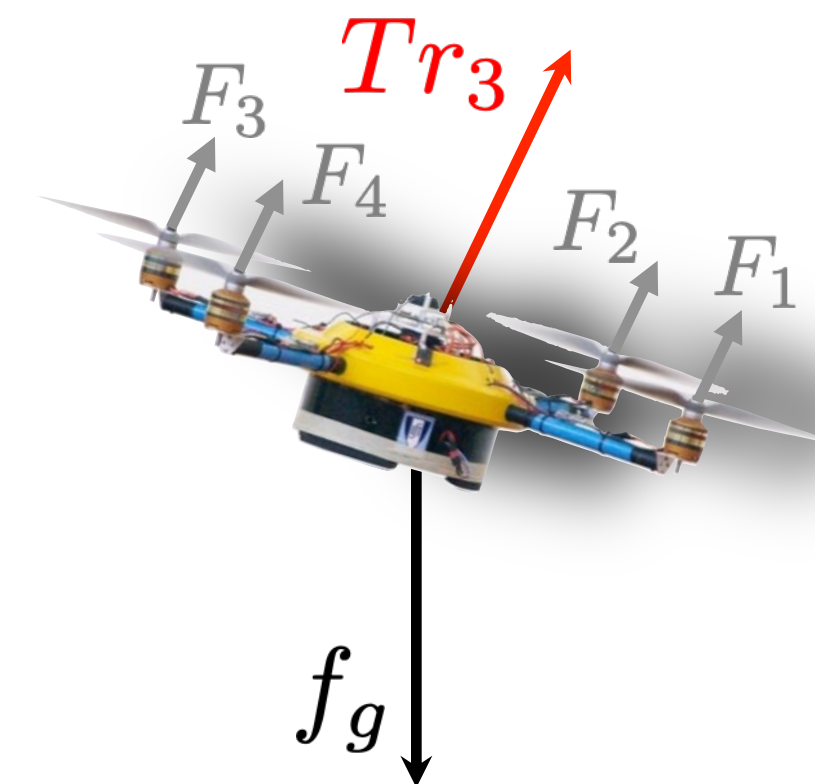
Desired output (p_d, ψ_d)

Quadrotor forces and moments

Two pairs of counter-rotating rotors



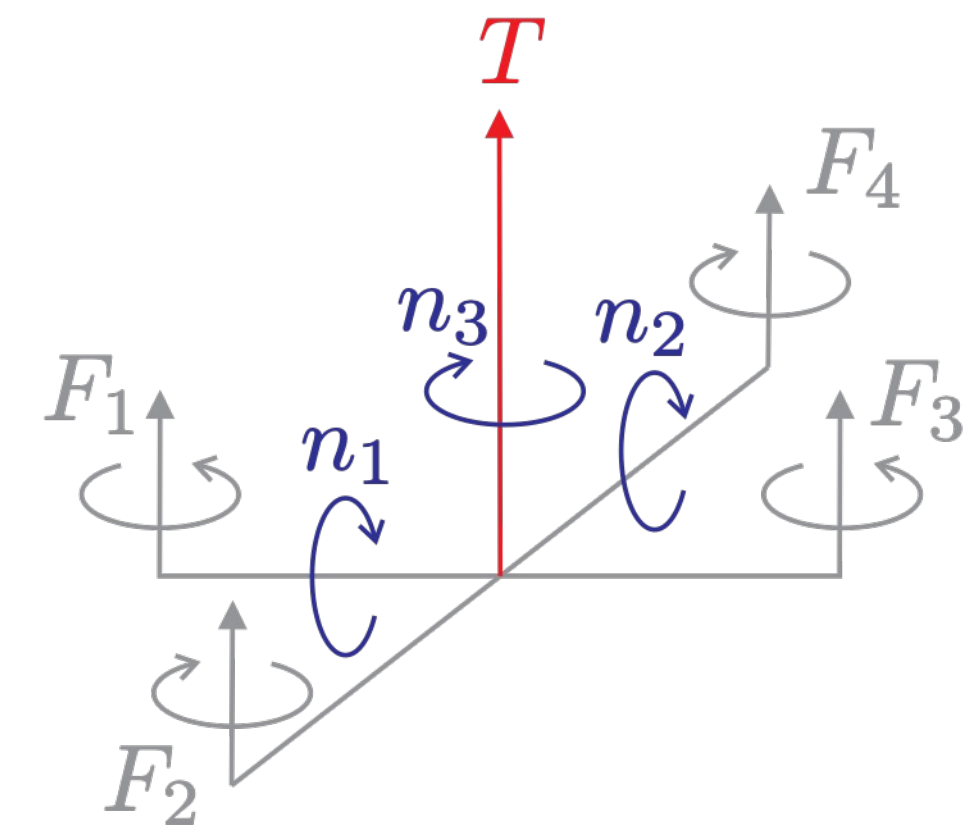
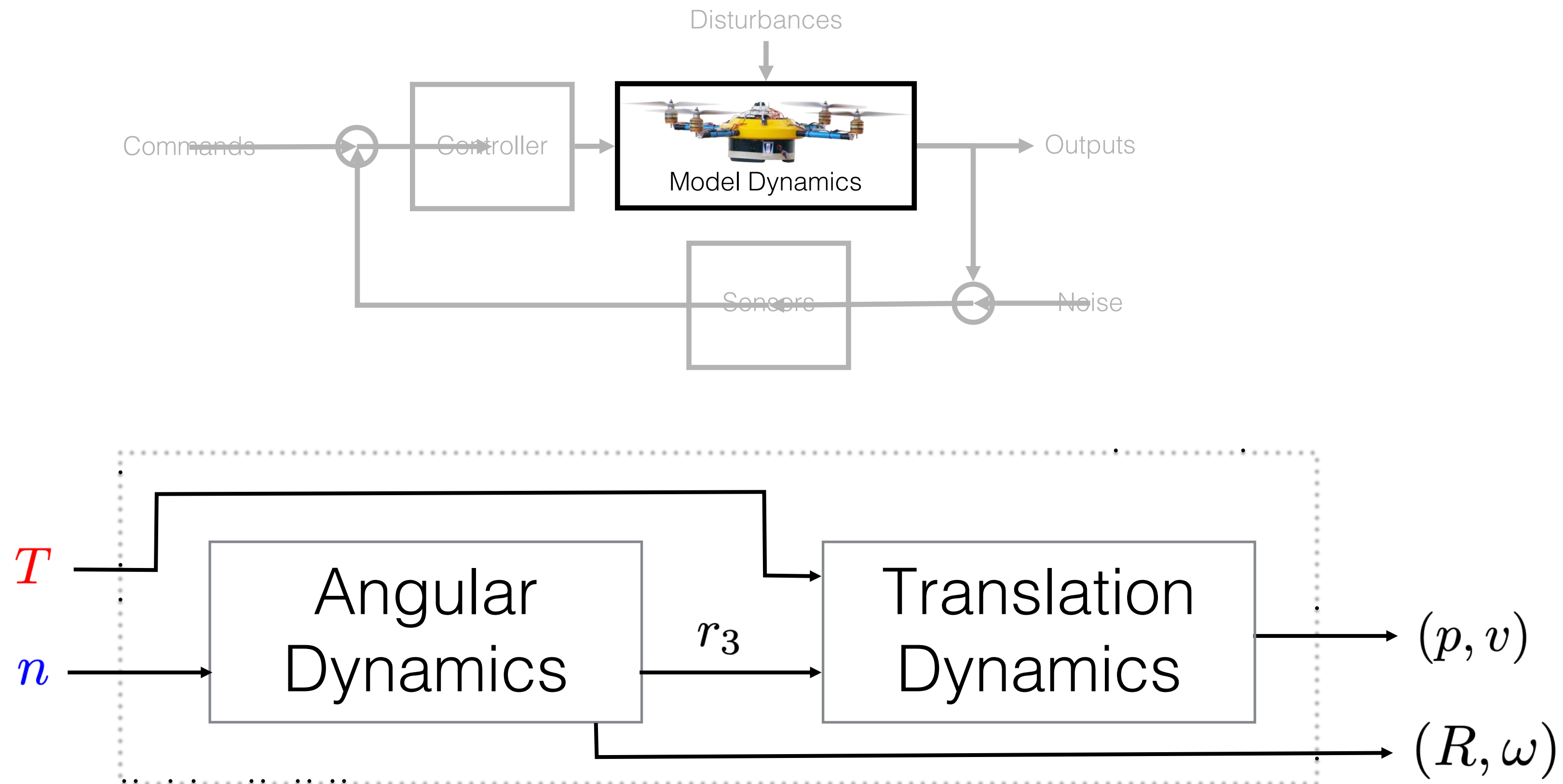
$$\begin{bmatrix} T \\ n_x \\ n_y \\ n_z \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & b & 0 & -b \\ -b & 0 & b & 0 \\ c & -c & c & -c \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}$$



$$r_3 = Re_3 = R \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$f_g = mge_3$$

Quadrotor dynamic model



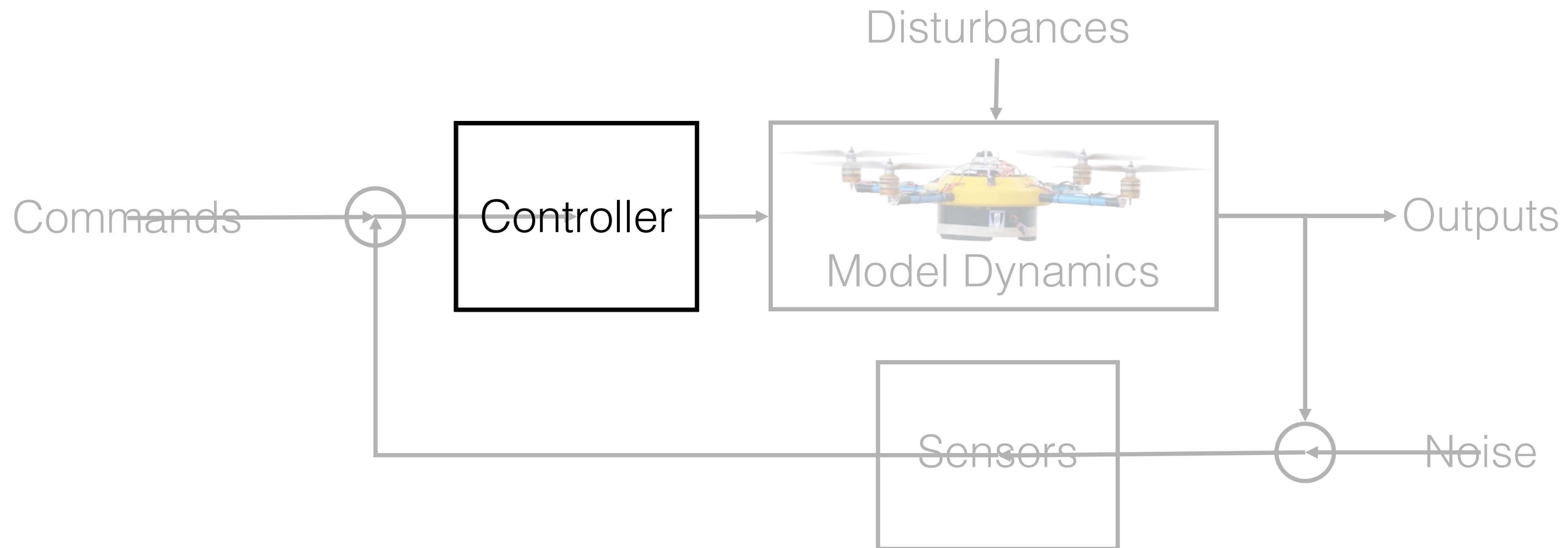
State (p, v, R, ω)

Input (T, n)

Underactuated system: 12 states, 4 inputs

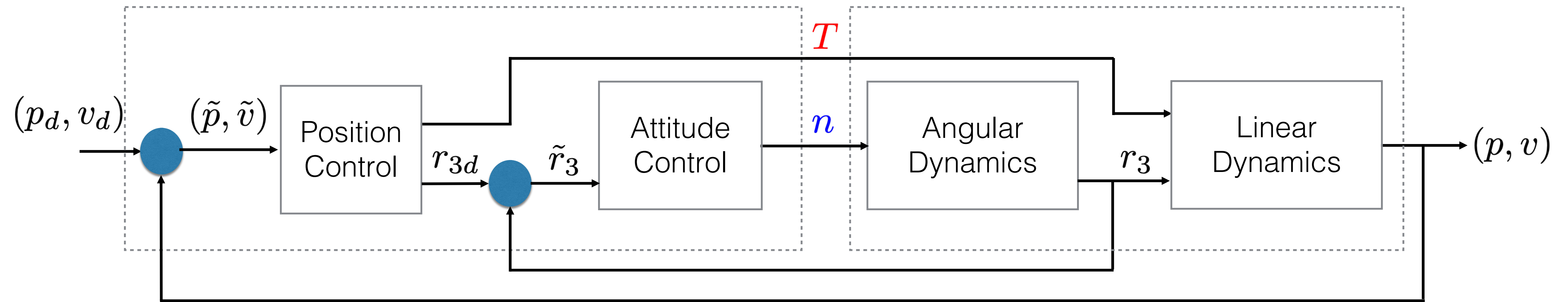
Can prescribe $(p, \psi) \in \mathbb{R}^4$

The quadrotor as a control system



Now the controller

Trajectory Tracking Control

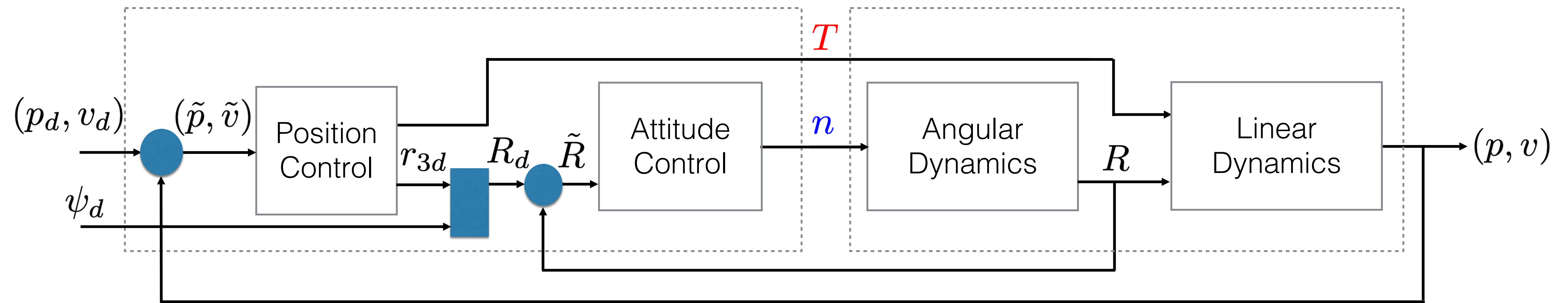


Simplest Position Controller yields a mass-spring-damper system

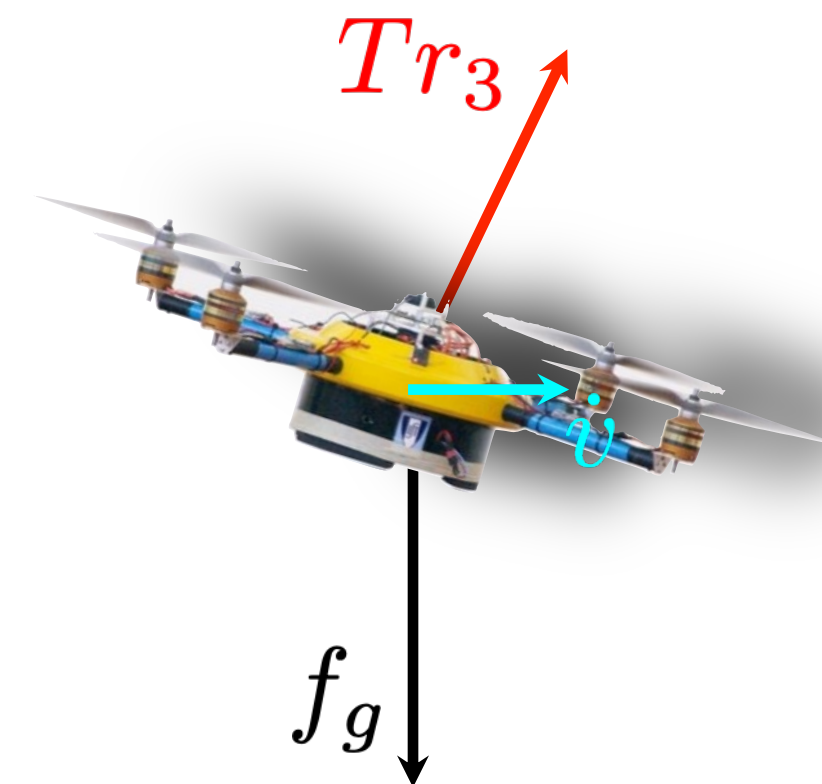
$$m\ddot{\tilde{v}} = -k_1\tilde{p} - k_2\tilde{v}$$



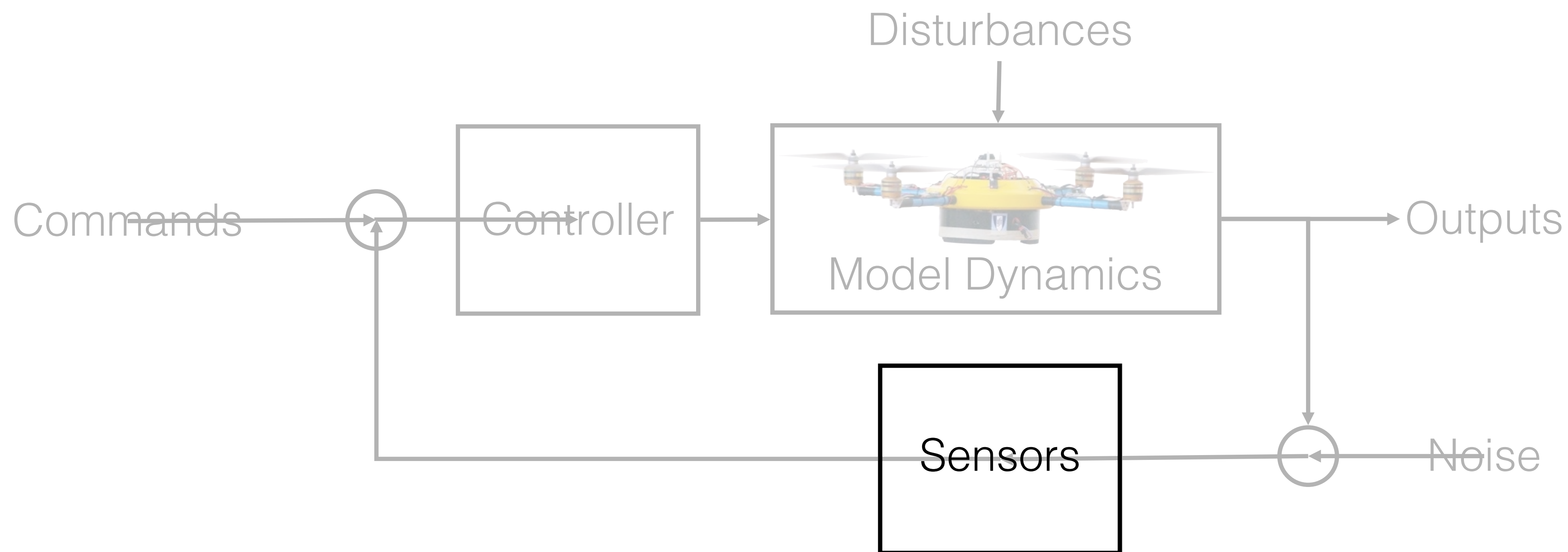
Trajectory Tracking Control



Desired position and yaw angle
 (p_d, ψ_d)



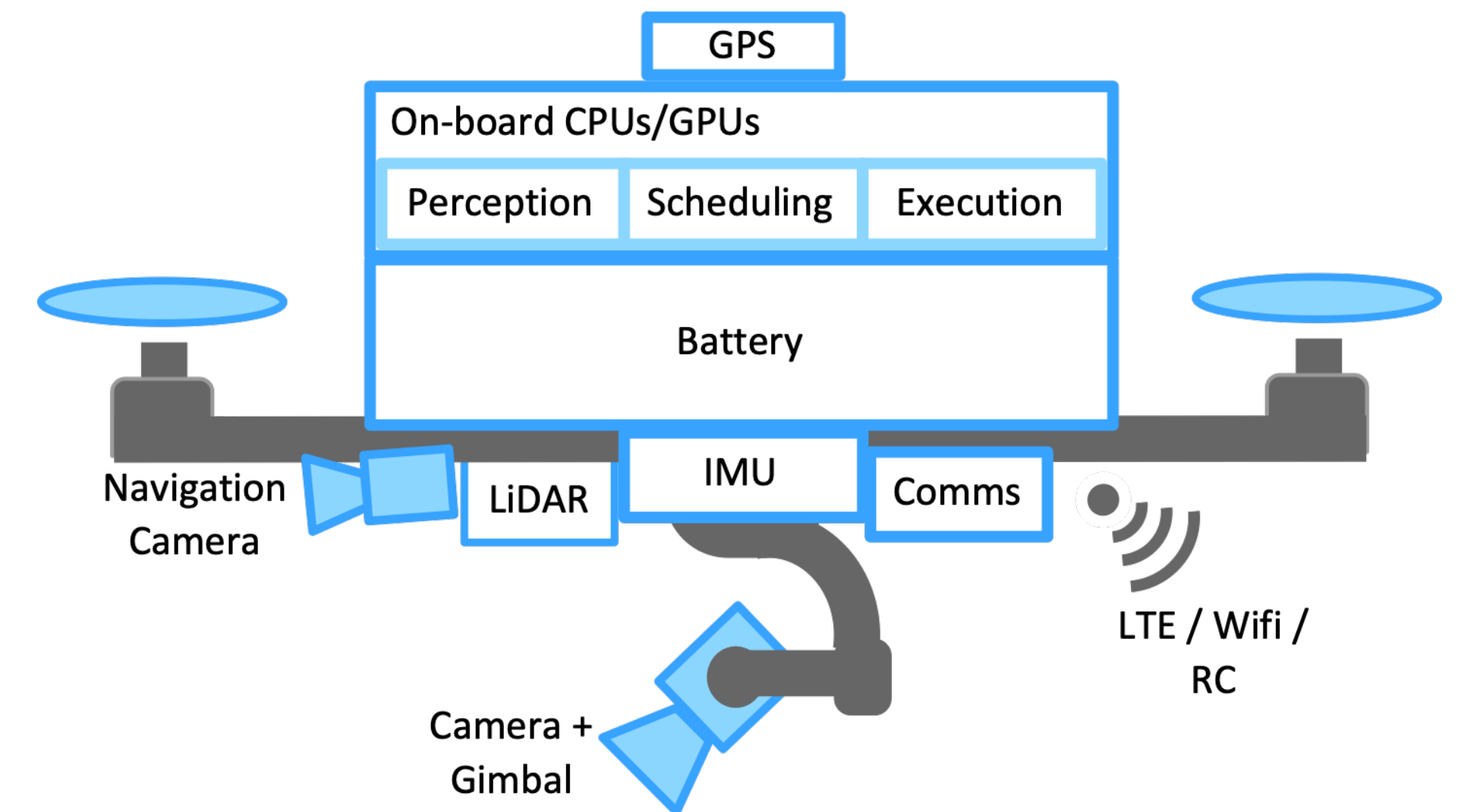
The quadrotor as a control system



Now the sensors

Sensors and Onboard Architecture

- Sensors
 - GPS - position estimates
 - Barometer or sonar : height estimate
 - IMU (accelerometers, gyroscopes, magnetometers) - attitude estimates
 - Cameras and LiDARs - mainly used for obstacle avoidance and task execution (surveying, inspection, ...)
- Onboard Computers
- Communication Systems
- Batteries



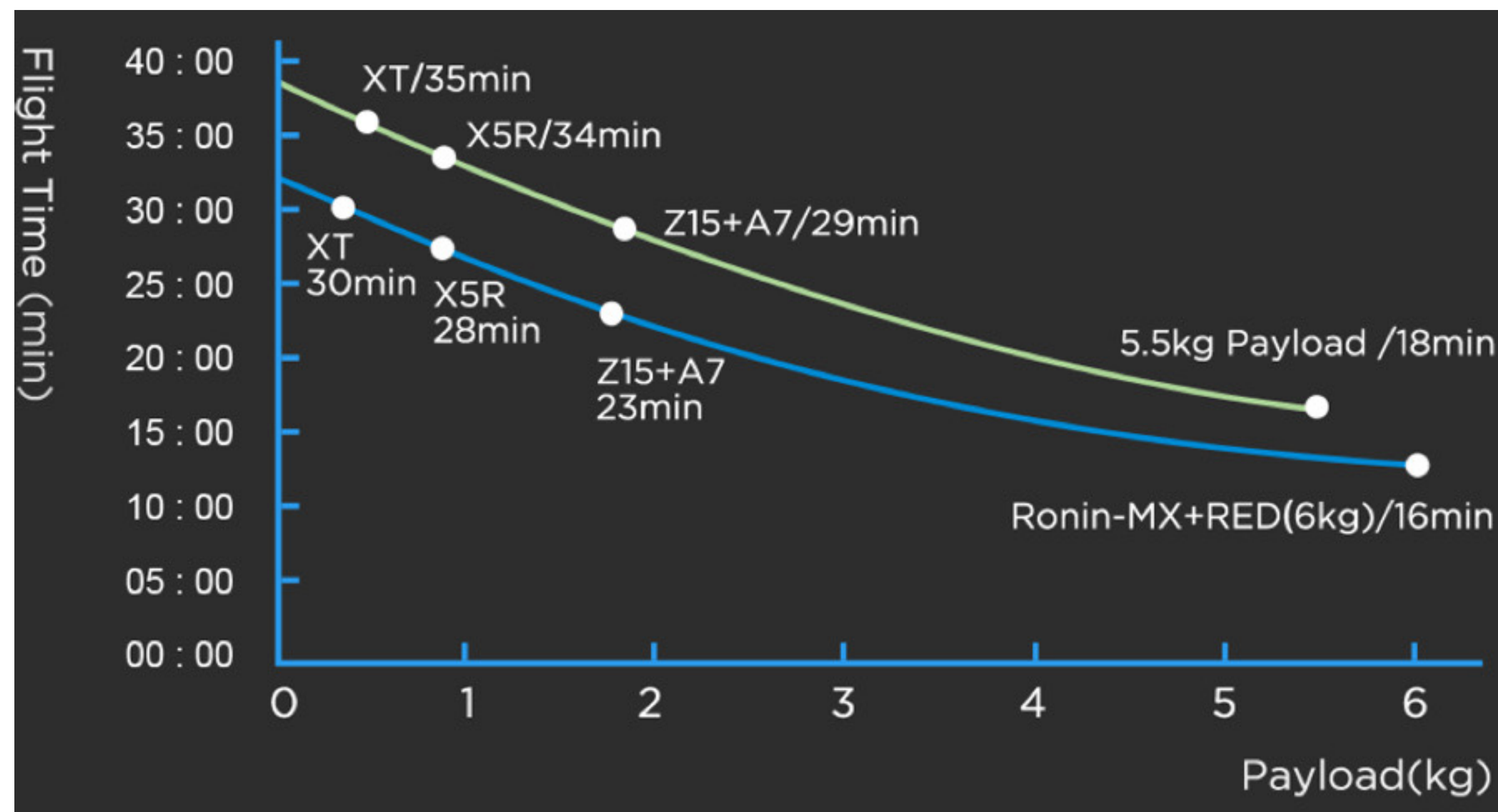
Biggest limitation to enhanced autonomy

- Flight time *vs.* available payload

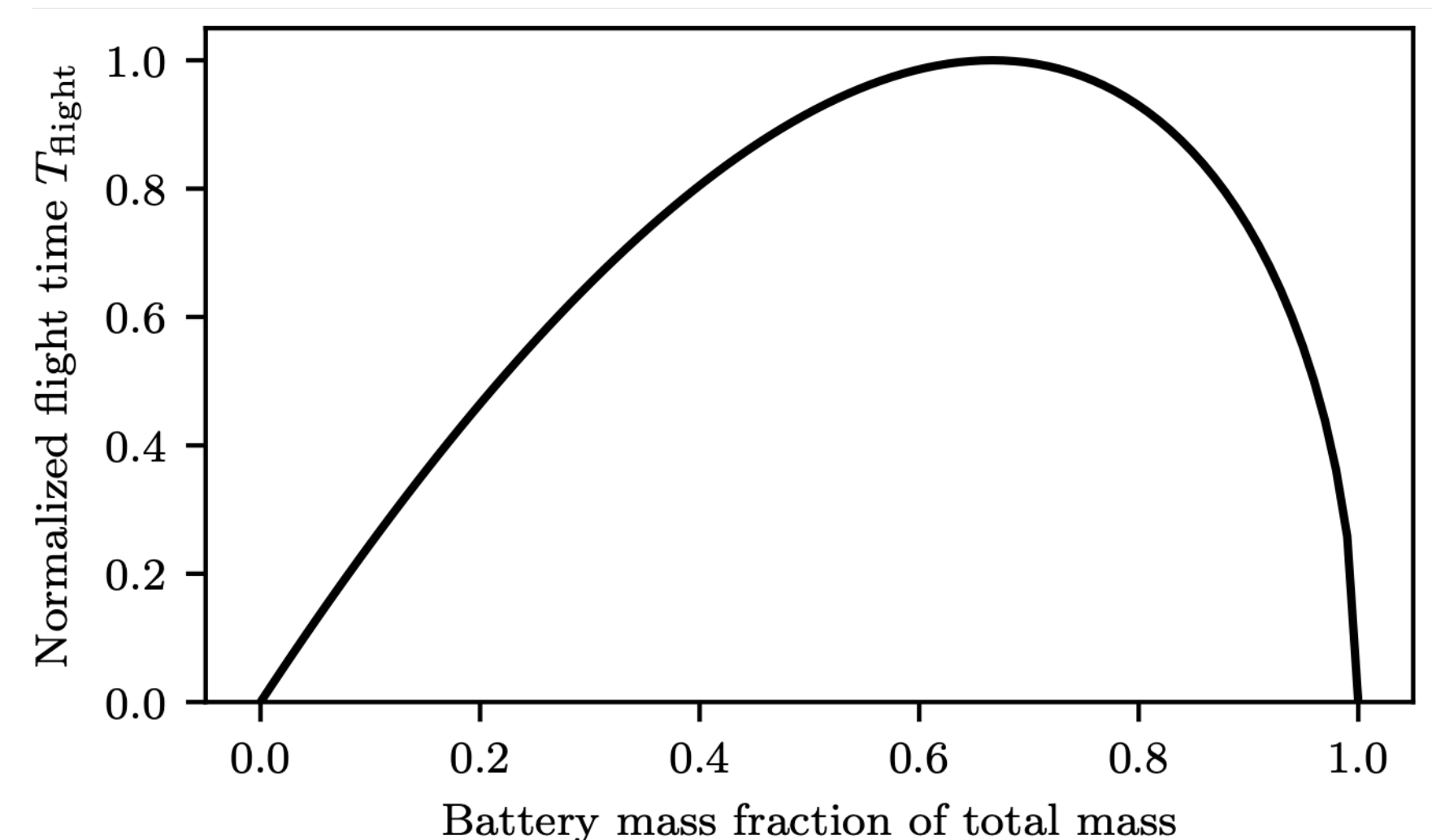
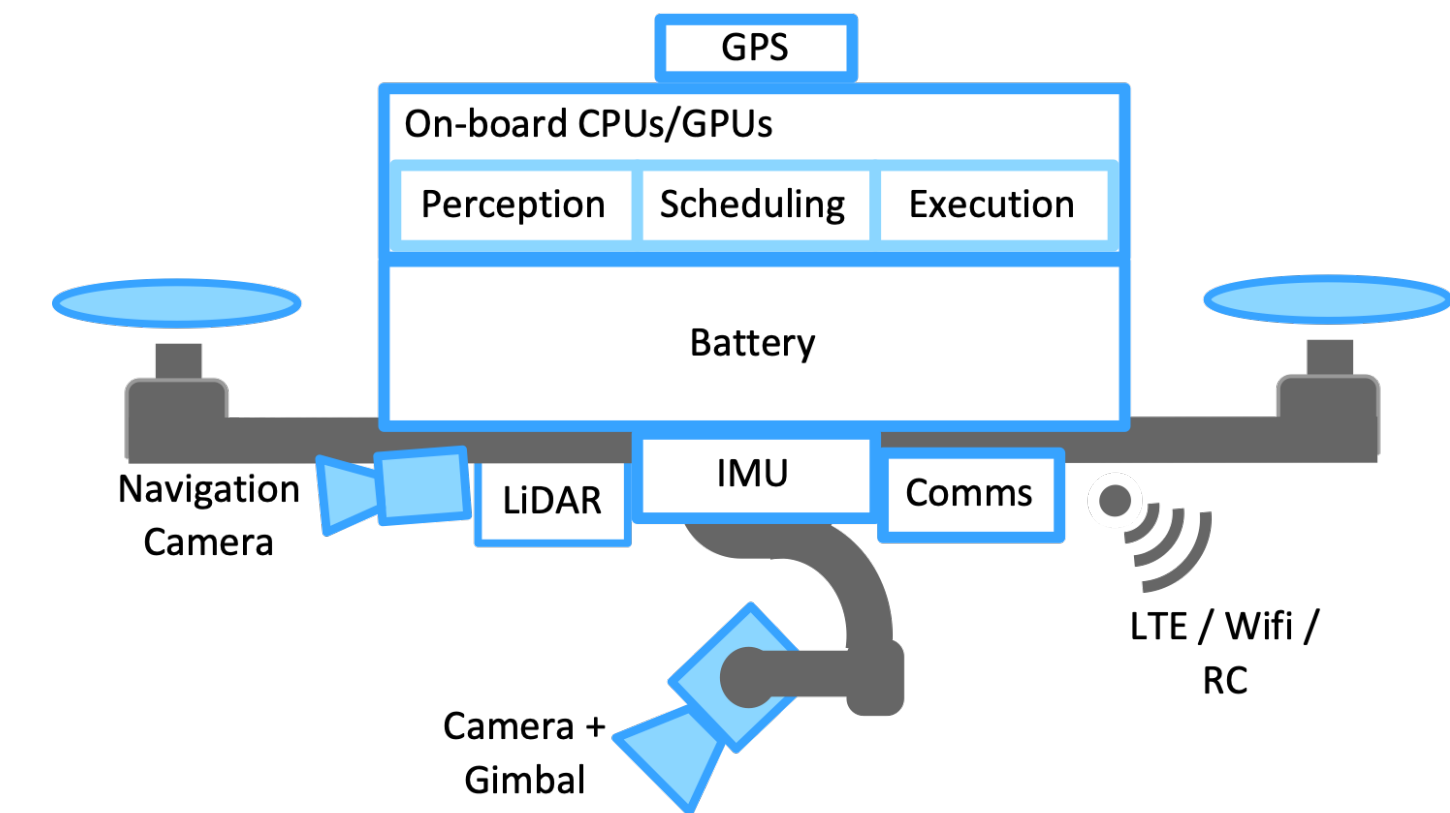
Onboard Architecture

Biggest limitation to enhanced autonomy

- Flight duration *vs.* available payload



Source: <https://www.dji.com/pt/matrice600-pro>

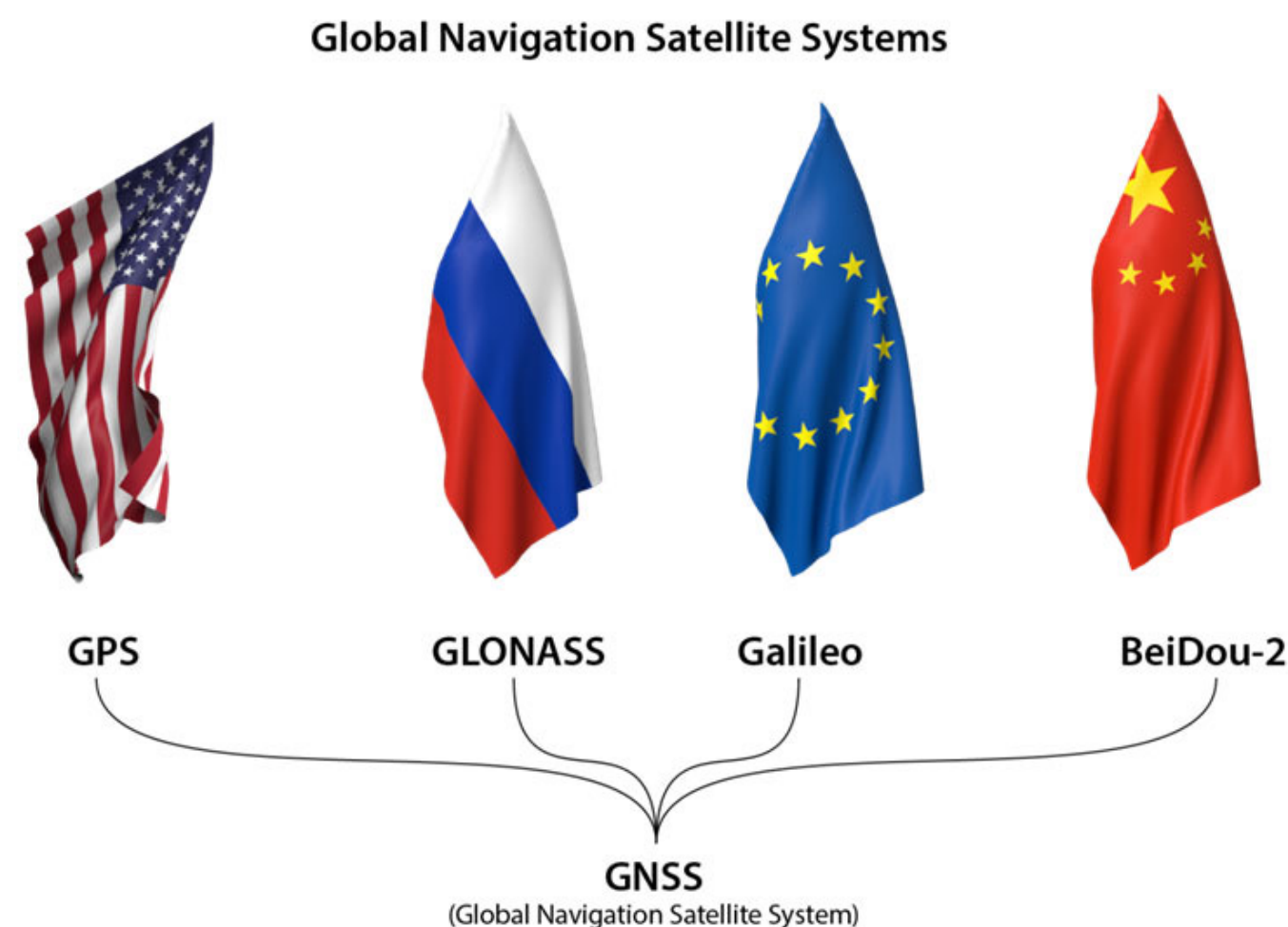


Source:

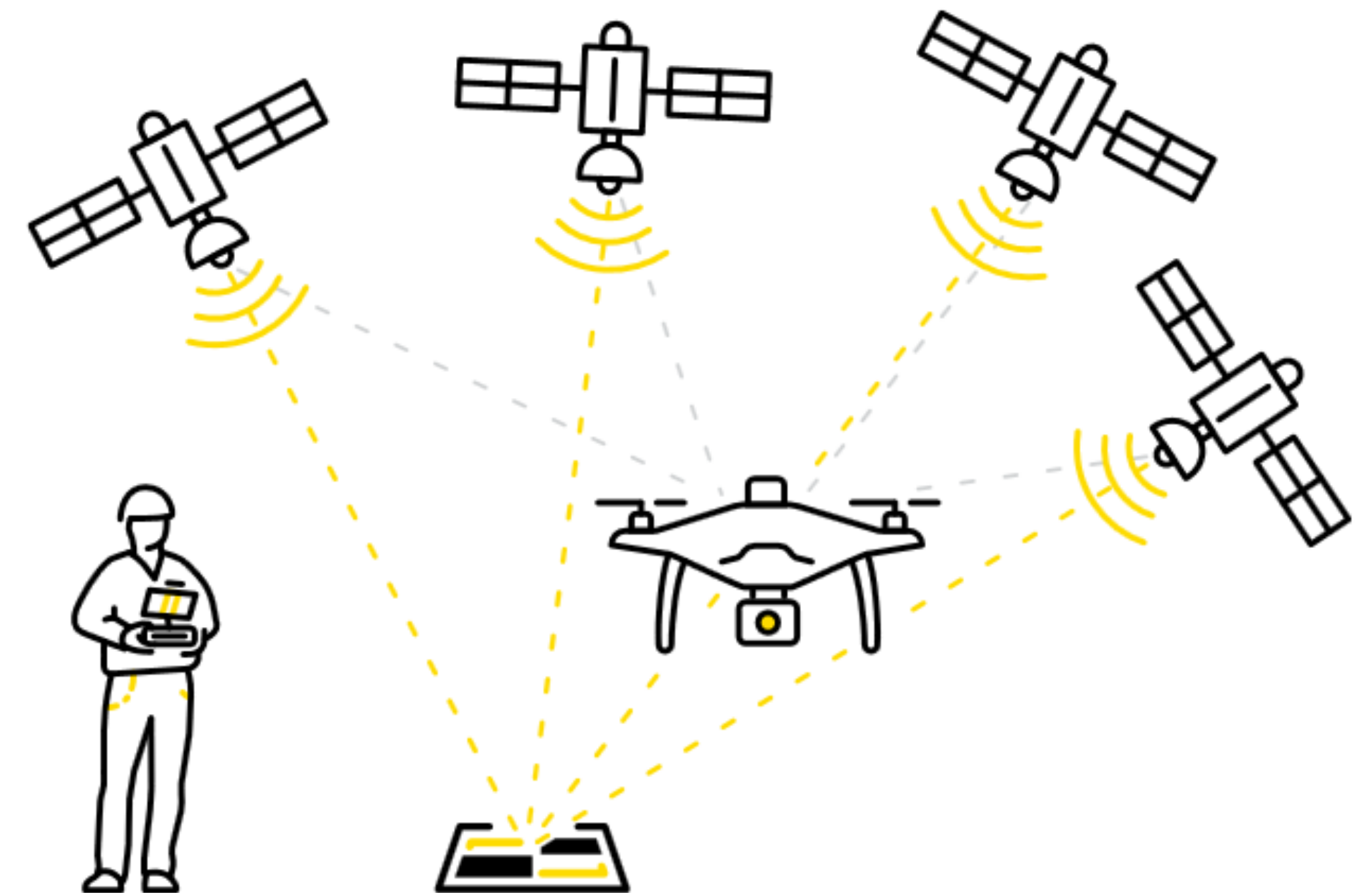
<https://spectrum.ieee.org/automan/robotics/drones/swappable-flying-batteries-keep-drones-aloft-almost-forever>

GNSS - Global Navigation Satellite System

- Used to estimate **position** based on time and known position of dedicated satellites.



+geostationary satellites

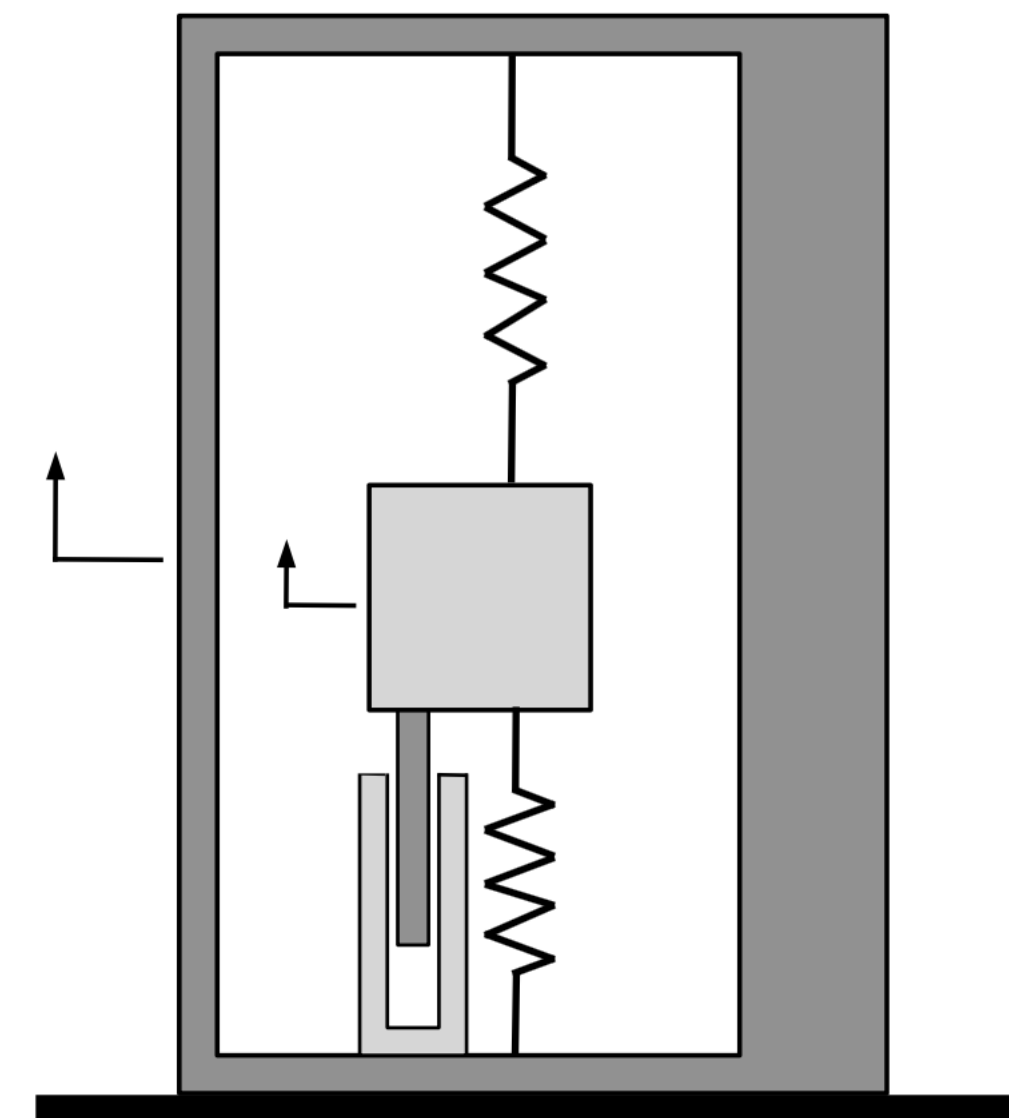
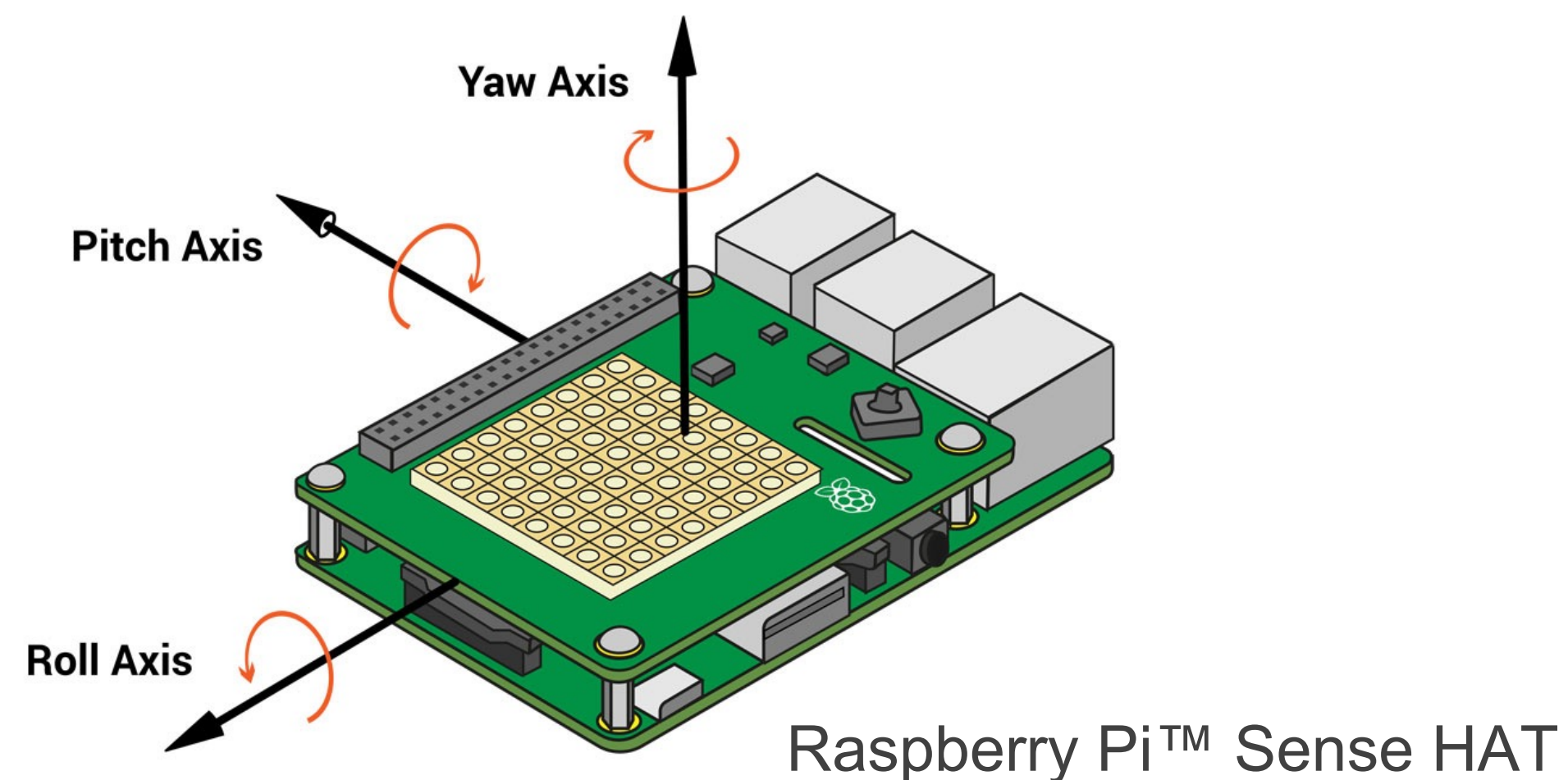


IMU - Inertial Measurement Unit

- Three-axis accelerometers, gyroscopes, and magnetometers
- Used to estimate the attitude of the vehicle

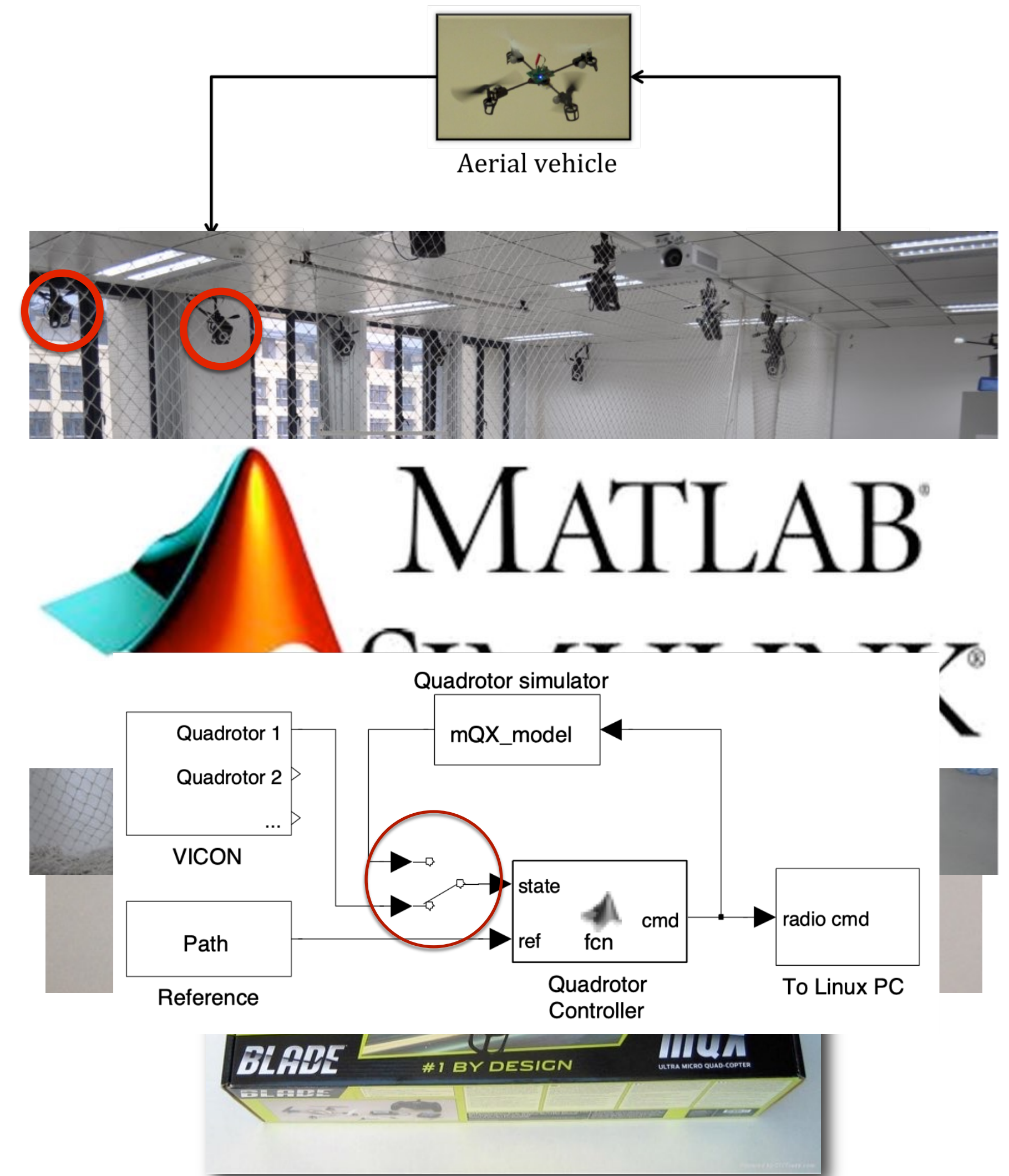
How a Smartphone Knows Up from Down (accelerometer)

<https://www.youtube.com/watch?v=KZVgKu6v808>

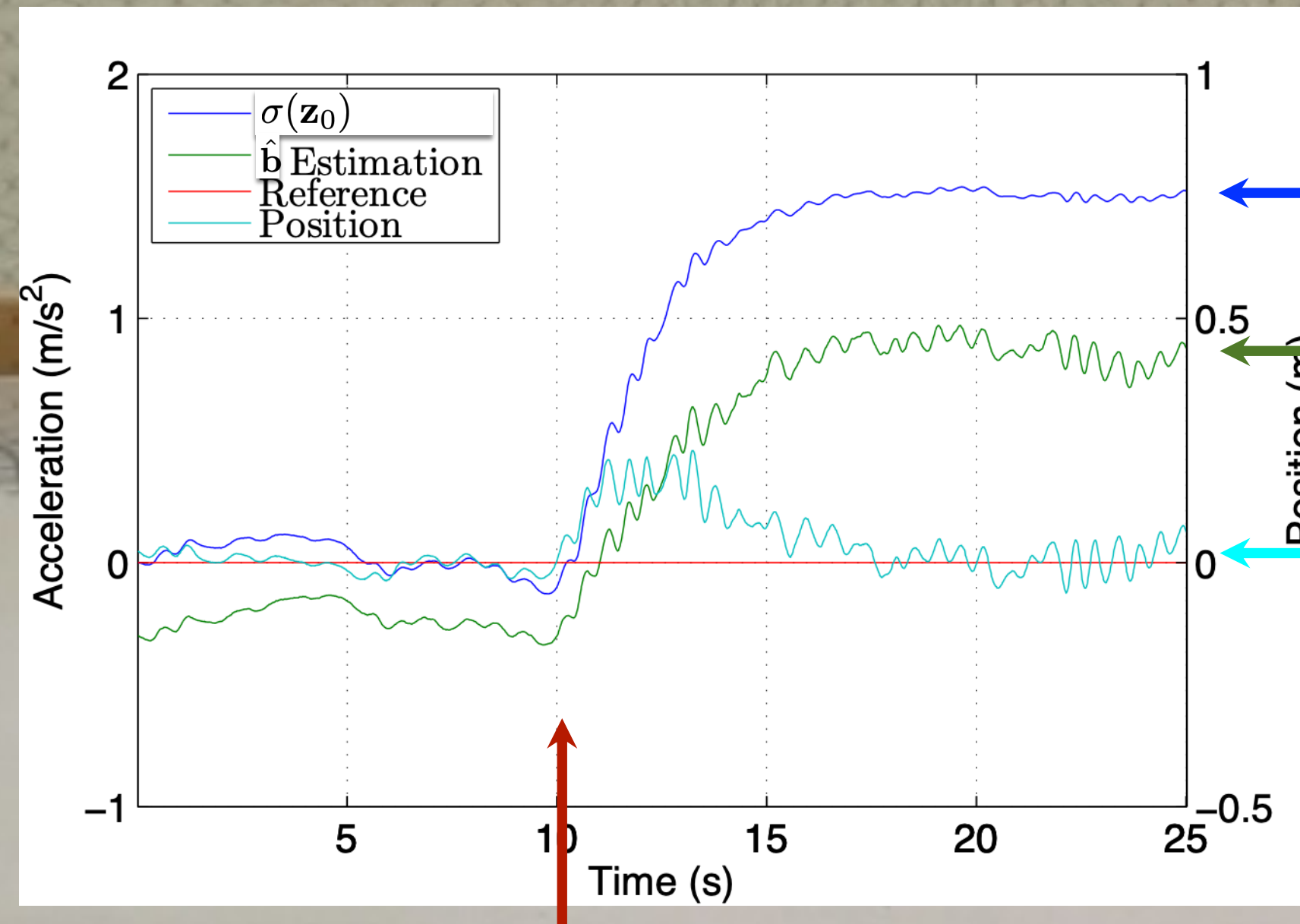


Indoors - Experimental Setup

- Readily available quadrotor solution
- Computer controlled radio transmitters
- Motion Capture system for state measurements:
→ **ARENA in ACCAII lab**
- Integrated system
- Rapid prototyping and experimental testing of controllers



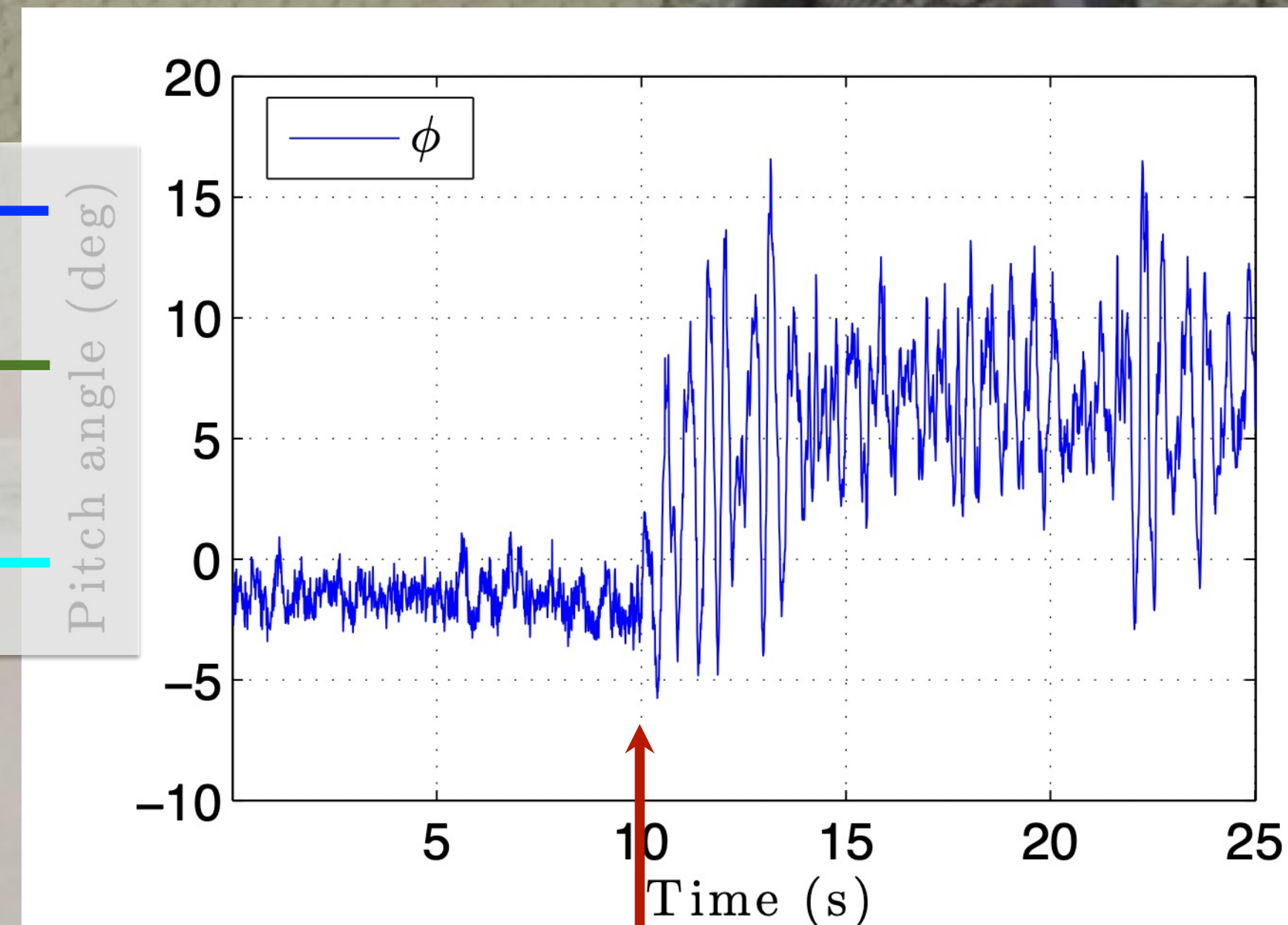
A few application examples in ISR lab



Fan ON

Position

Estimation and
integral action



Fan ON



<http://users.isr.ist.utl.pt/~rmac/videos/aggressive.mp4>



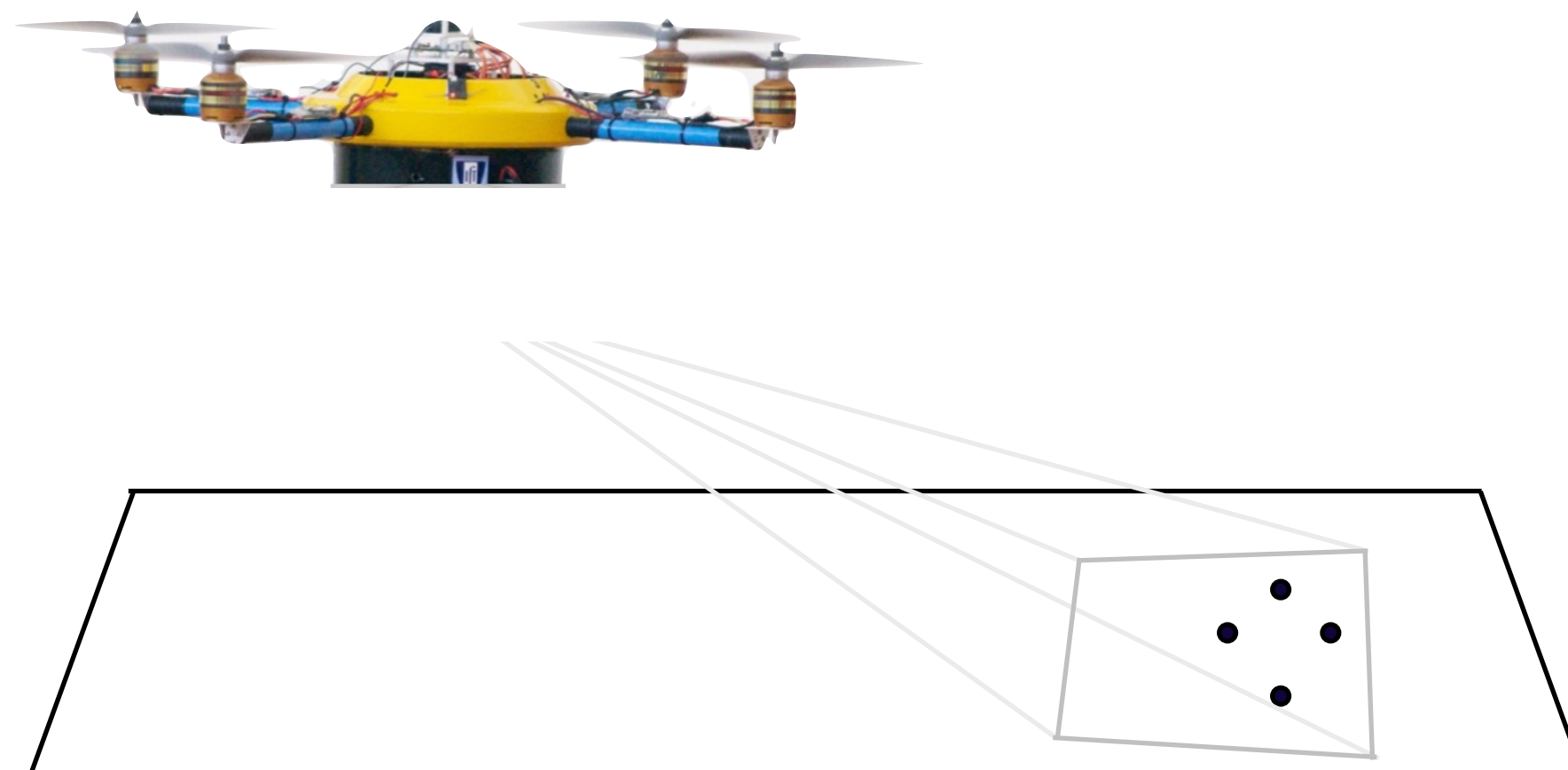
Landing on a moving platform using vision-based control

Landing on a moving platform

- Quadrotor
- IMU
- Wide-angle camera
- *Goal:* land on a target

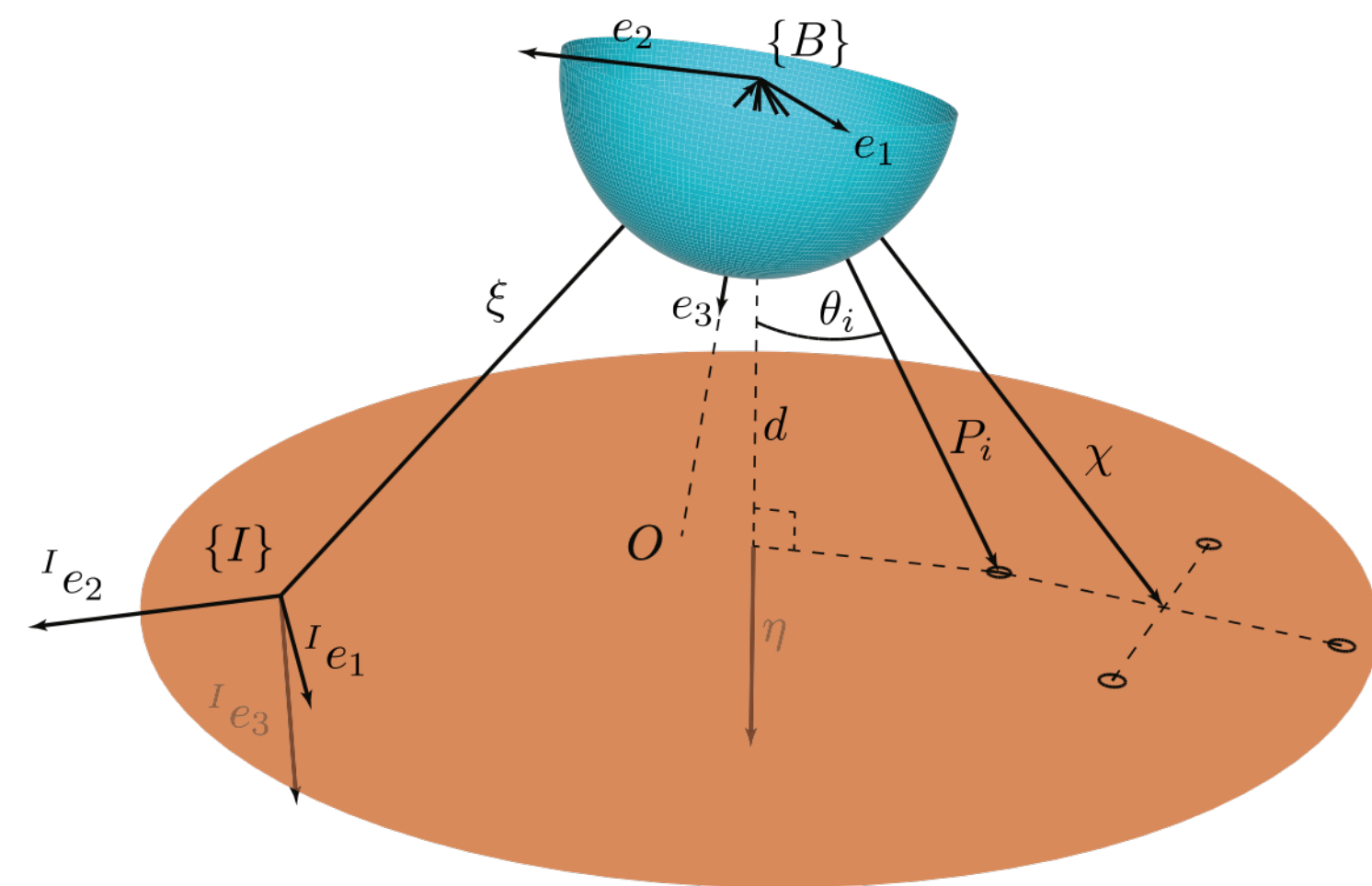
Assumptions

- Planar target
- Moving platform
 - Translation
- Non-aggressive maneuvers



Vision-based controller

Control objective: $\chi \rightarrow 0$

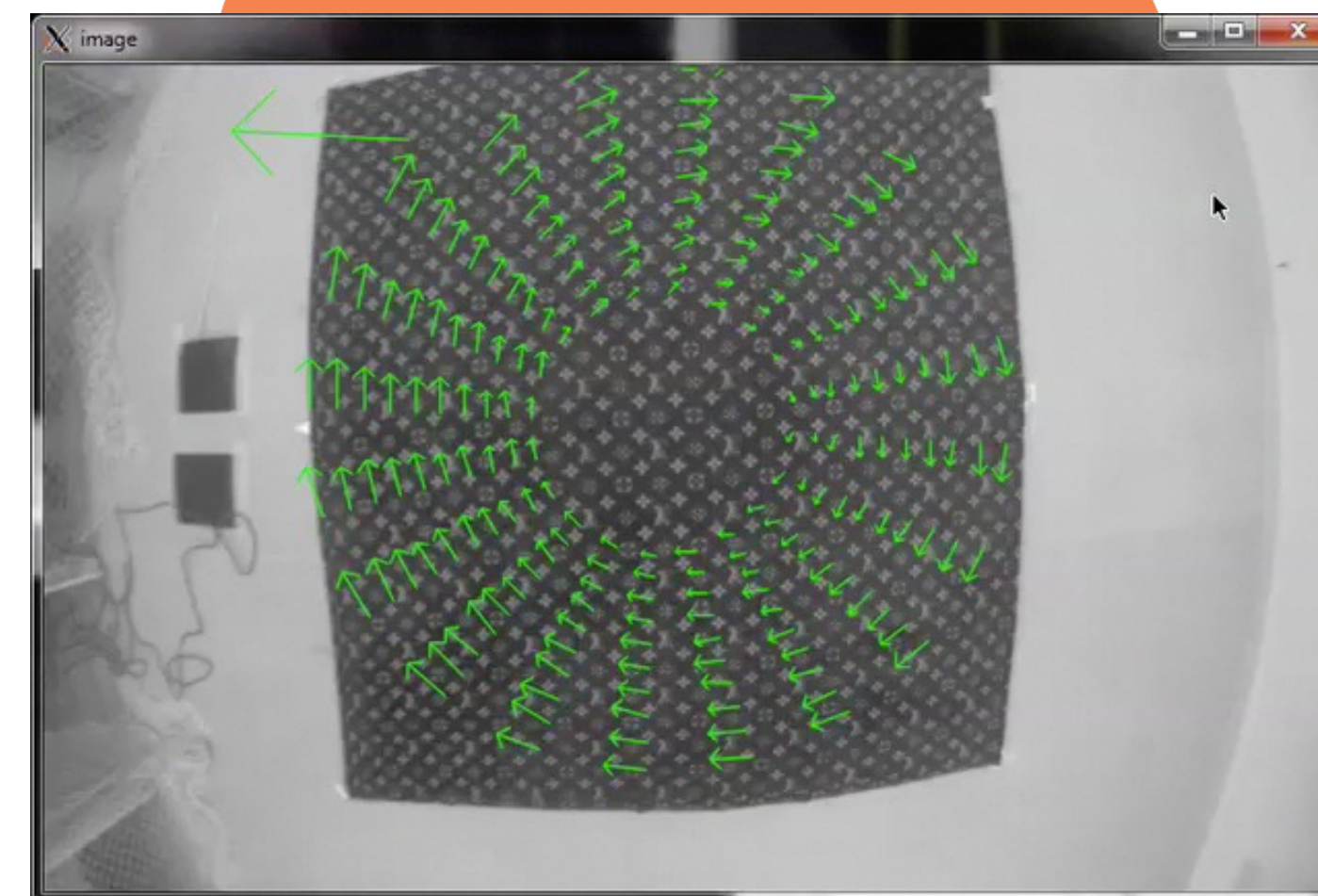
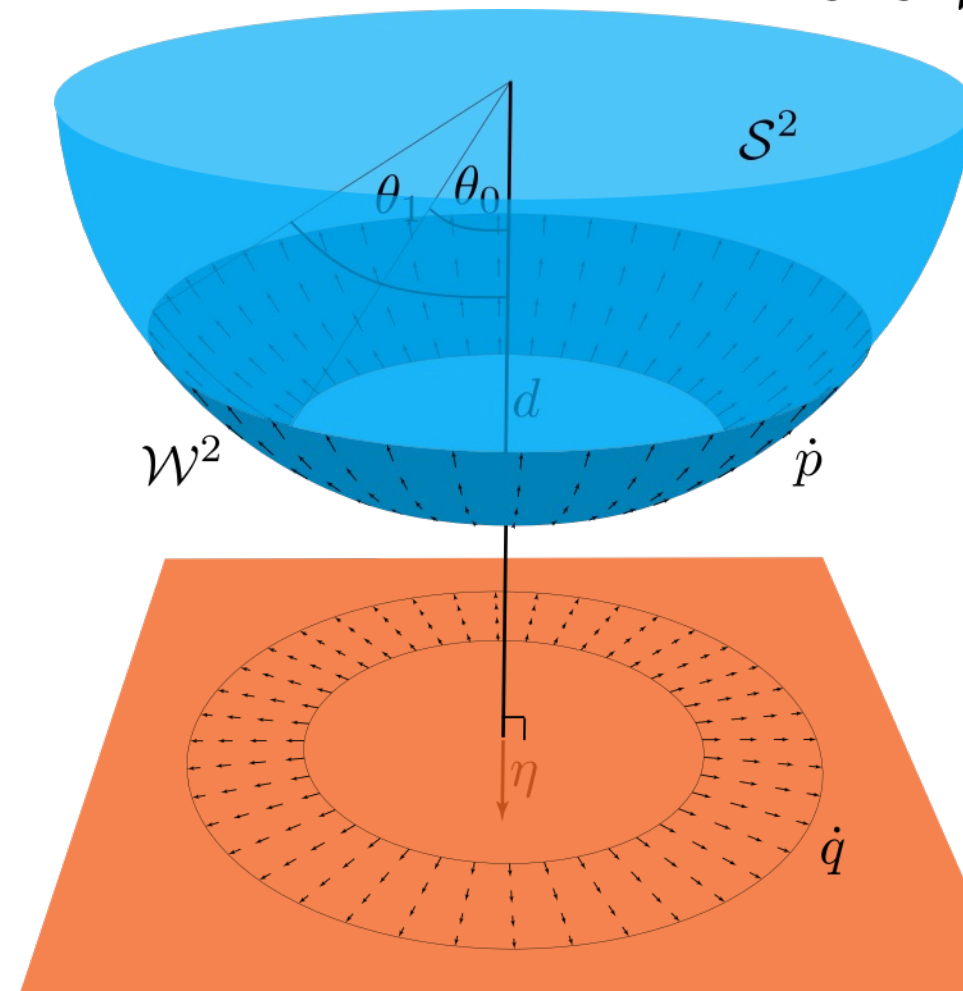


Available measurements:

- Images of target points

$$p_i = \frac{P_i}{\|P_i\|} \rightarrow \frac{P_i}{d} \rightarrow \frac{\chi}{d}$$

- Optical flow field $\phi = \iint_{\mathcal{W}^2} \dot{p} \, dp \rightarrow \frac{\dot{\chi}}{d}$



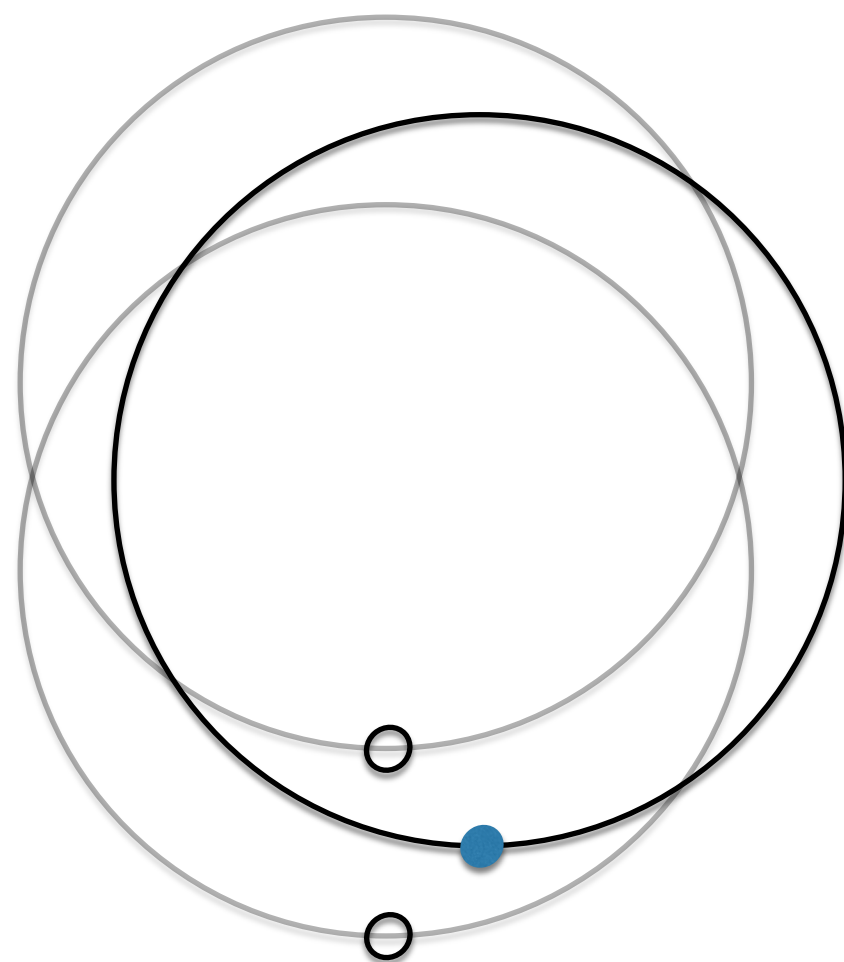




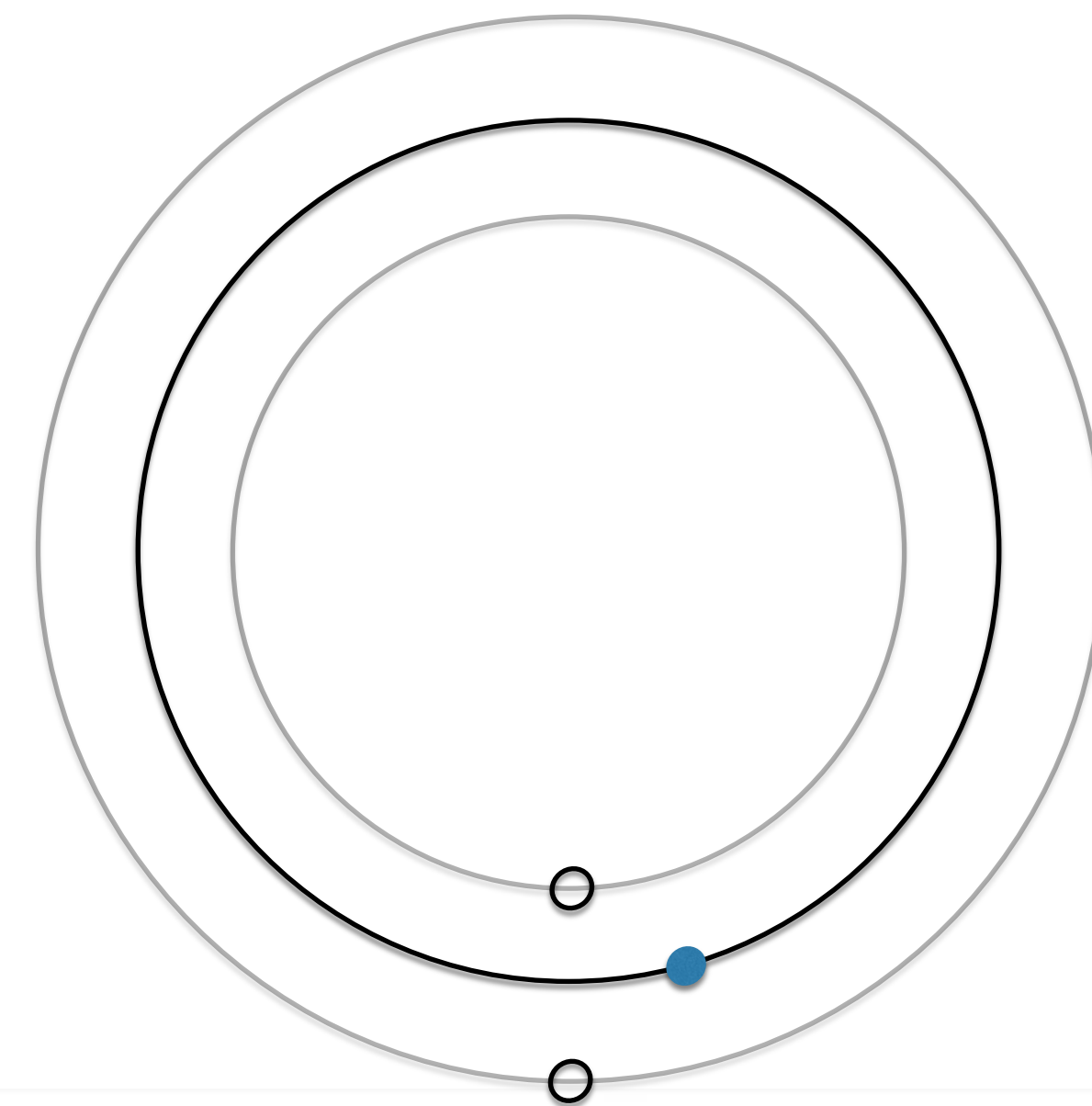
Leader-following for formation control

Leader-following for formation control

- Main idea:
Trailer-like behavior for the followers.



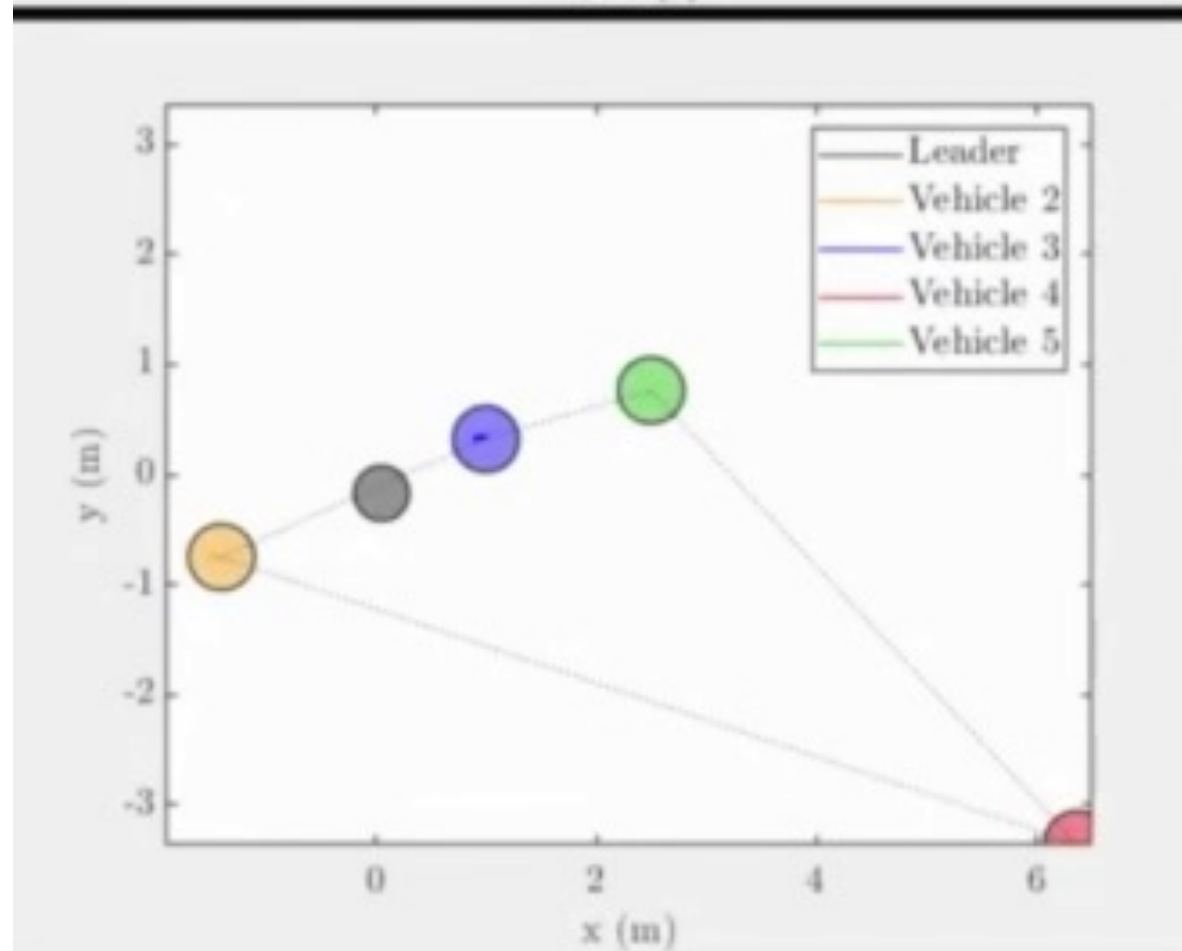
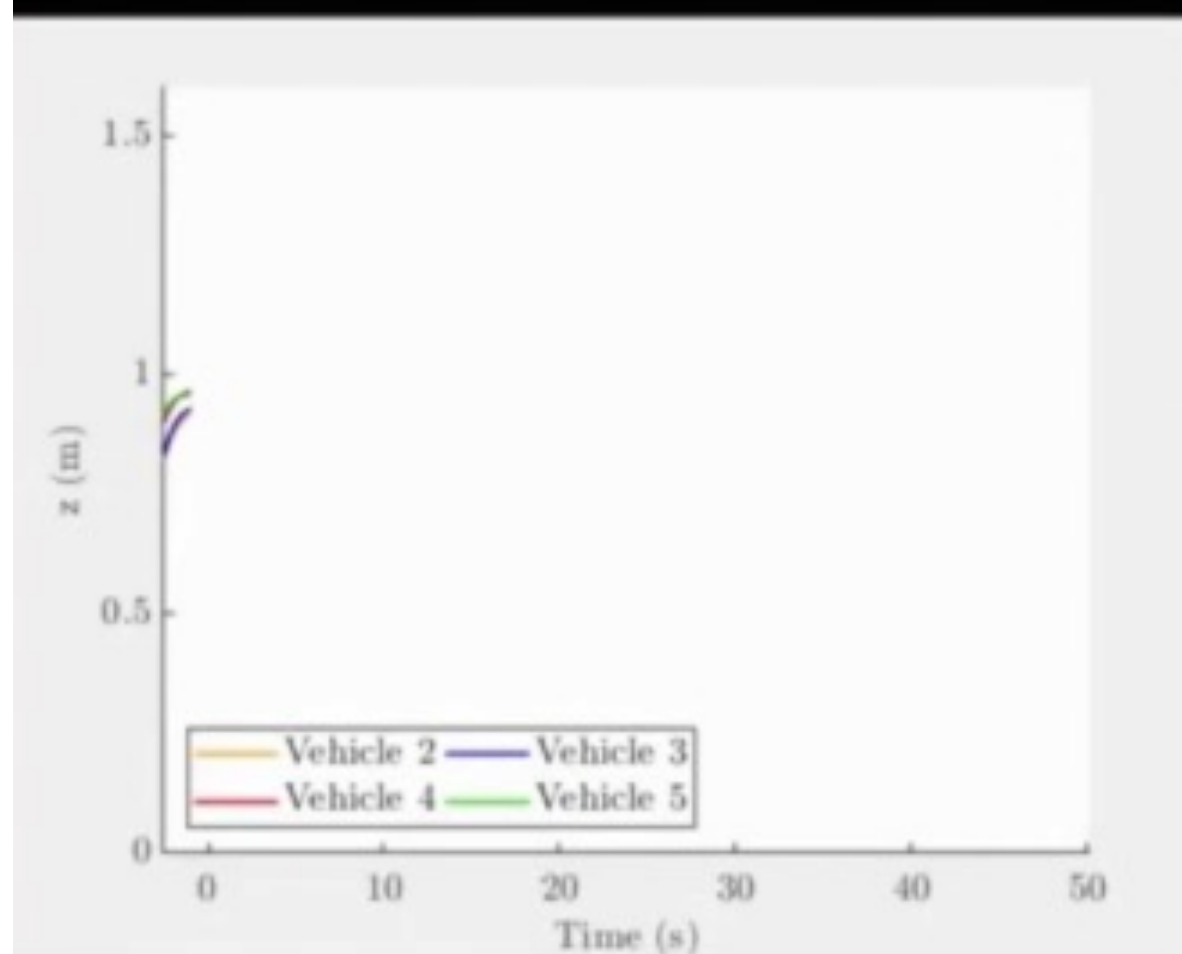
In inertial frame:
Translated identical paths



In trailer frame:
Different paths, no superposition



http://users.isr.ist.utl.pt/~rmac/videos/multi_vehicles.mp4

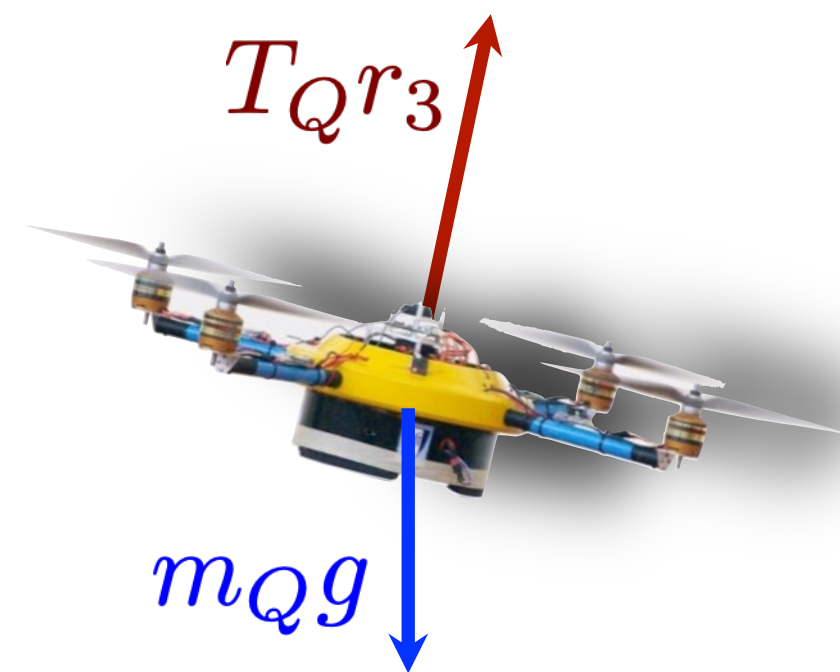




Slung-load transportation using quadrotors

Modeling

Quadrotor



State

$$(p_Q, v_Q, R_Q, \omega_Q)$$

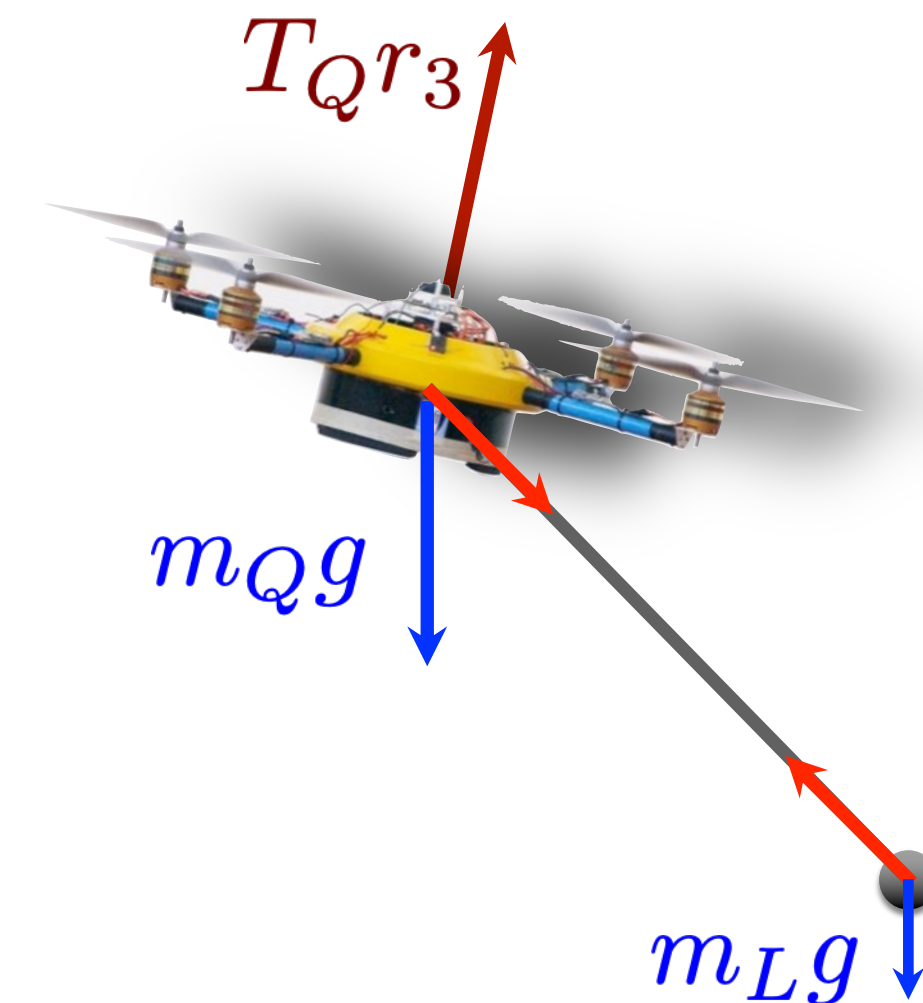
Input

$$(T_Q, n_Q) \in \mathbb{R}^4$$

Flat output

$$(p_Q, \psi_Q) \in \mathbb{R}^4$$

Quadrotor + Load



State

$$(p_Q, v_Q, R_Q, \omega_Q, p_L, v_L)$$

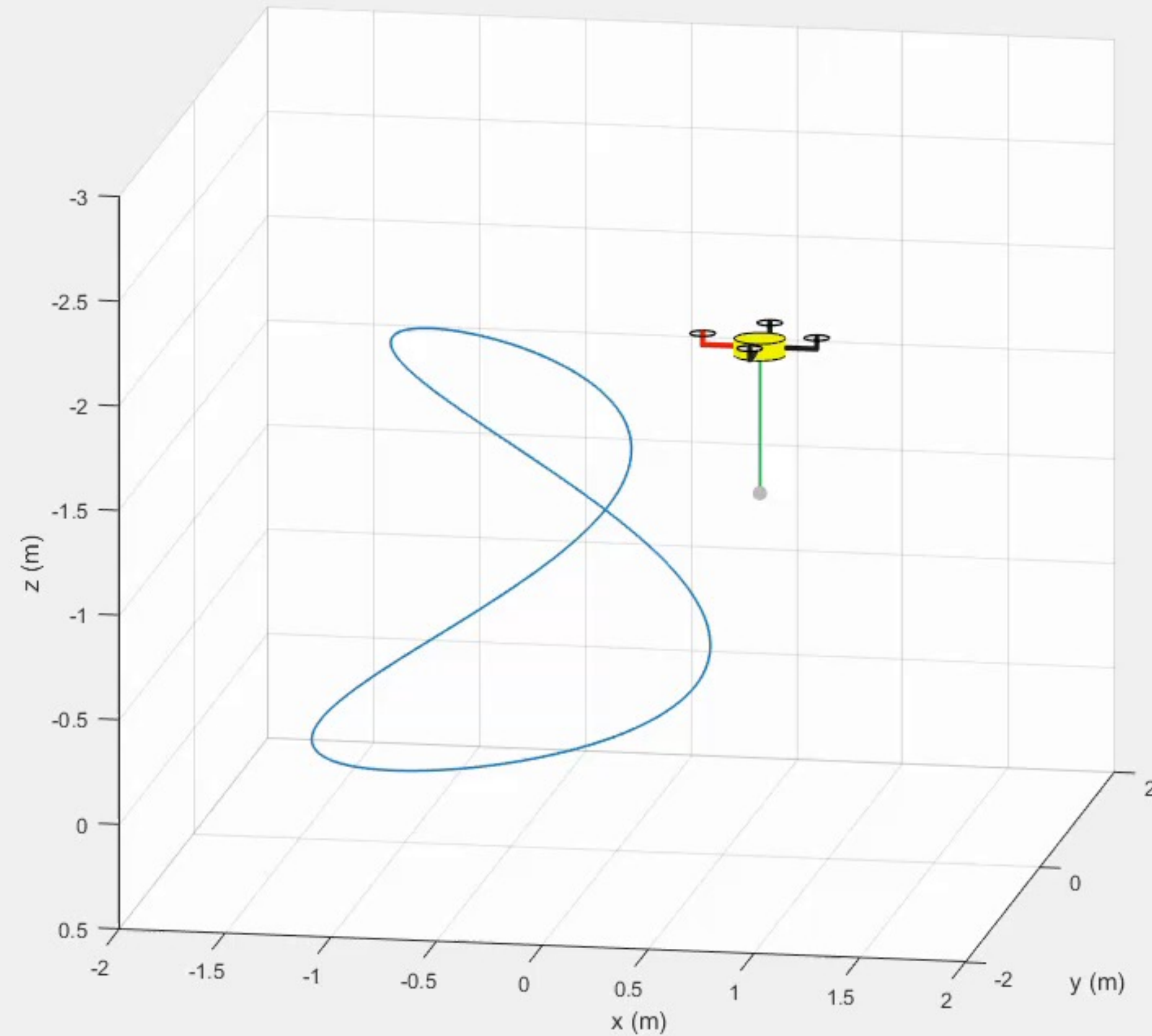
Input

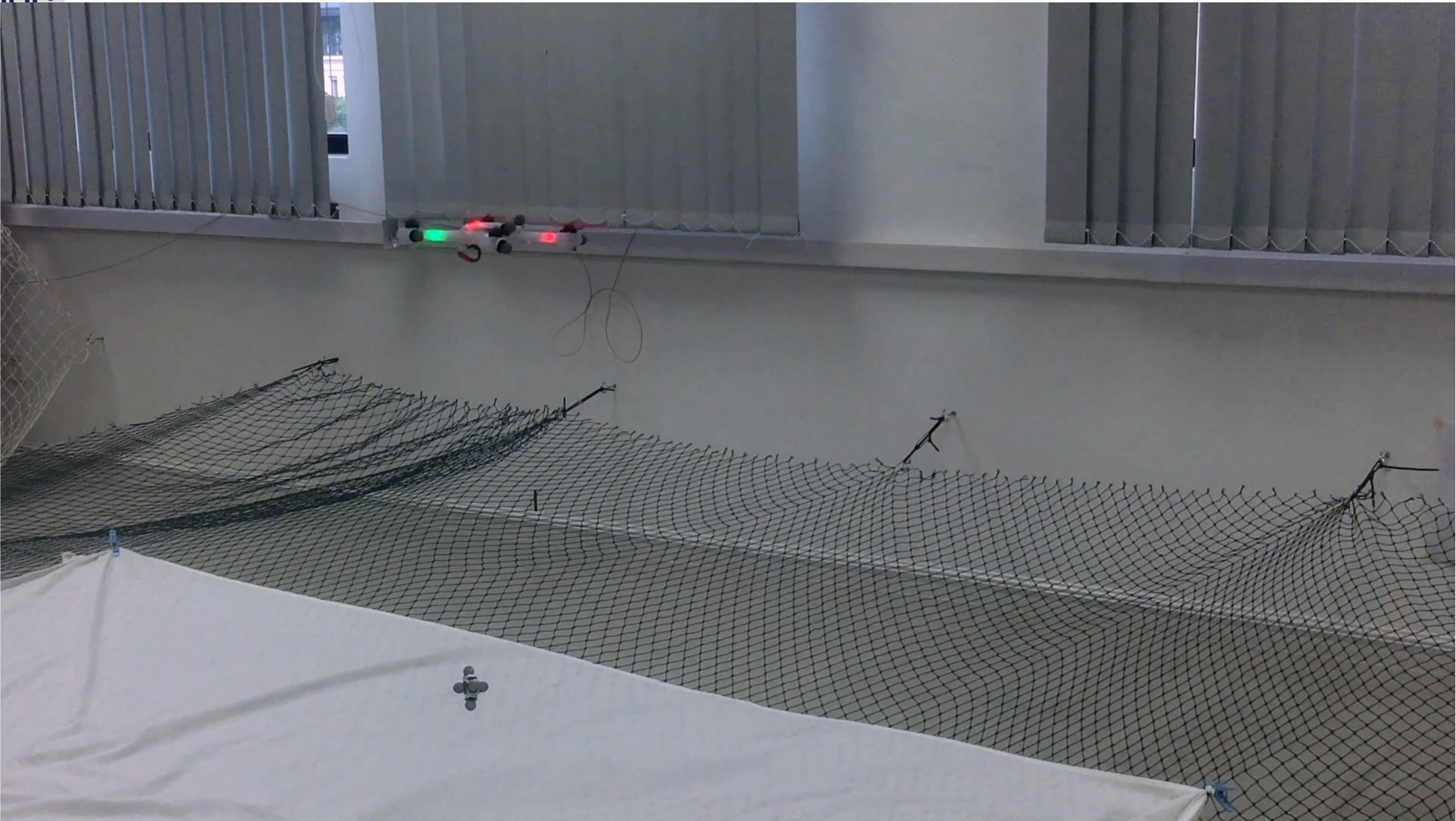
$$(T_Q, n_Q) \in \mathbb{R}^4$$

Flat output

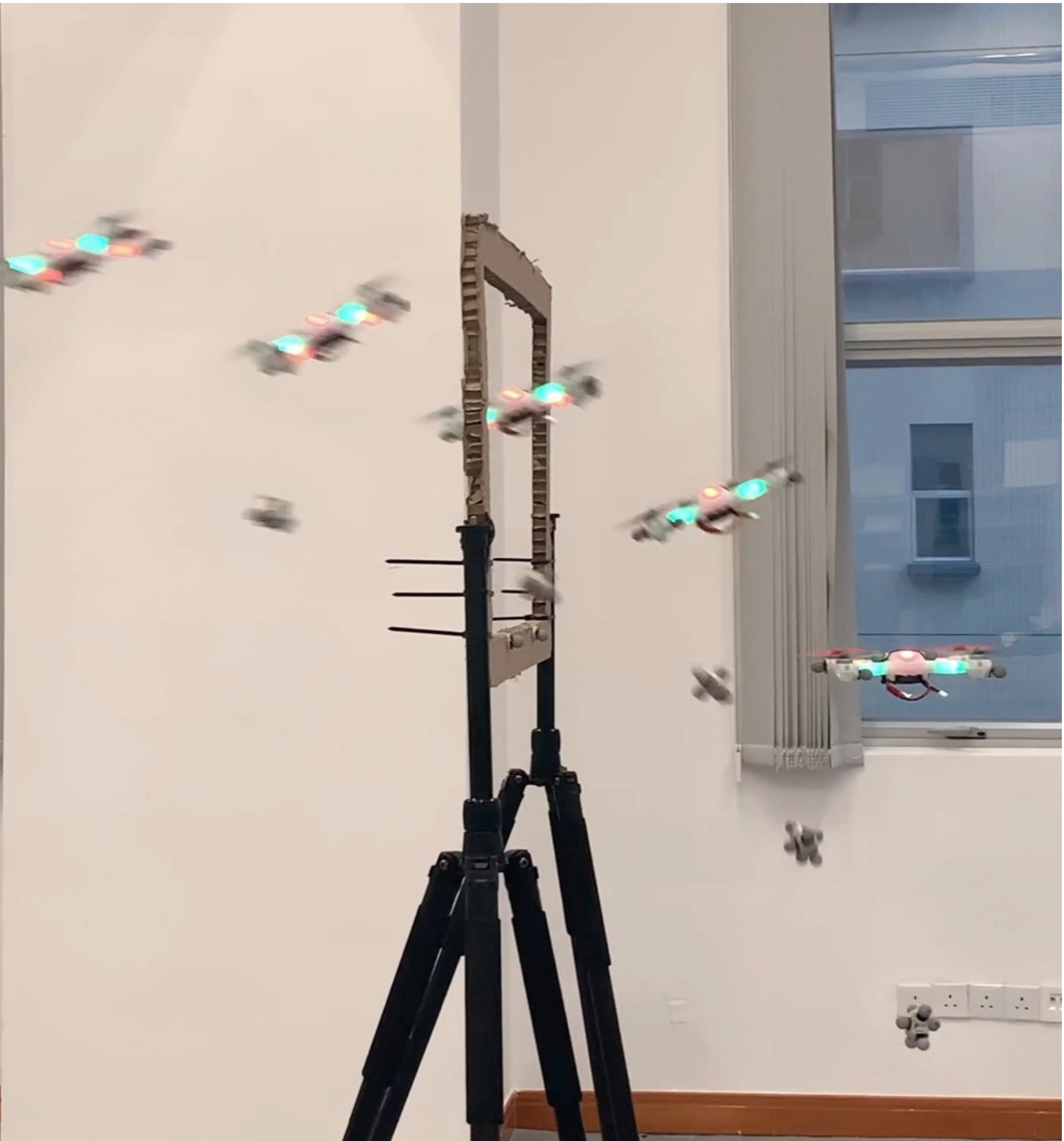
$$(p_L, \psi_Q) \in \mathbb{R}^4$$

Simulation results





http://users.isr.ist.utl.pt/~rmac/videos/load_transportation.mp4



Acknowledgments

The team (current and past members)

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